You Asked

Section A

1. (a) For each of the following four networks, discuss the consequences if a connection fails : (3)

(i) Five devices arranged in a mesh topology

P.T.O.

(ii) Five devices arranged in a bus topology

(iii) Five devices arranged in a ring topology

(b) For each of the following network applications, indicate whether TCP or UDP would best fit the service requirements of the application, and justify. (3)

(i) Multiplayer online first-person shooting game

(ii) Text-based instant messaging app

(iii) Video Streaming

(c) Calculate the maximum data rate of a channel with a bandwidth of 200KHz when sixteen levels of digital signaling are used. (3)

Gemini 3.1 Pro

Here are the solutions to the networking questions presented in the image:

1. (a) Consequences of a Connection Failure

(i) Five devices arranged in a mesh topology: In a fully connected mesh topology, every device has a dedicated point-to-point link to every other device. If a single connection fails, it only impacts the two devices directly connected by that specific link. The rest of the network remains entirely unaffected, and the two impacted devices can still communicate by routing their data through the other devices in the network.

(ii) Five devices arranged in a bus topology: This depends on which connection fails:

- **Backbone failure:** If the main central cable (the bus) breaks, the entire network goes down. The break causes signals to bounce back and create interference, halting all communication.
- **Drop-line failure:** If the cable connecting a single device to the main backbone fails, only that specific device is disconnected. The rest of the network continues to function normally.

(iii) Five devices arranged in a ring topology: In a standard, unidirectional ring topology, data travels in one direction from device to device. If a single connection (link) between any two devices fails, the entire network fails because the loop is broken, and data/tokens can no longer circulate to reach their destinations.

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1. (b) TCP vs. UDP for Network Applications

(i) Multiplayer online first-person shooting game: UDP

- **Justification:** Fast-paced multiplayer games require the lowest possible latency to ensure real-time responsiveness. UDP does not waste time on error-checking, handshakes, or retransmitting lost packets. If a packet containing a player's coordinate is dropped, it's better to just receive the next updated coordinate immediately rather than halting the game to wait for a retransmission (which causes noticeable lag).

(ii) Text-based instant messaging app: TCP

- **Justification:** Text messaging requires complete reliability and in-order delivery. TCP ensures that every character of the message arrives intact and in the correct sequence. The slight delay introduced by TCP's error-checking and packet retransmission is completely unnoticeable in text communication, making its reliability the priority.

(iii) Video Streaming: UDP (for live/real-time) or **TCP** (for buffered VOD)

- **Justification:** For live video broadcasts, **UDP** is typically preferred because maintaining a continuous stream of frames is more important than perfect quality. A dropped packet might result in a momentary pixelation glitch, whereas TCP would pause the video to retrieve the lost packet, causing annoying buffering. *(Note: Modern Video on Demand (VOD) services like Netflix often use TCP because they pre-buffer the video, allowing time for retransmissions without pausing playback).* ---

1. (c) Maximum Data Rate Calculation

To calculate the maximum data rate of a noiseless channel, we use Nyquist's Bit Rate formula:

$$C = 2 \times B \times \log_2(L)$$

Where:

- C is the maximum data rate (capacity) in bits per second (bps)
- B is the bandwidth of the channel
- L is the number of signal levels used to represent data

Given values:

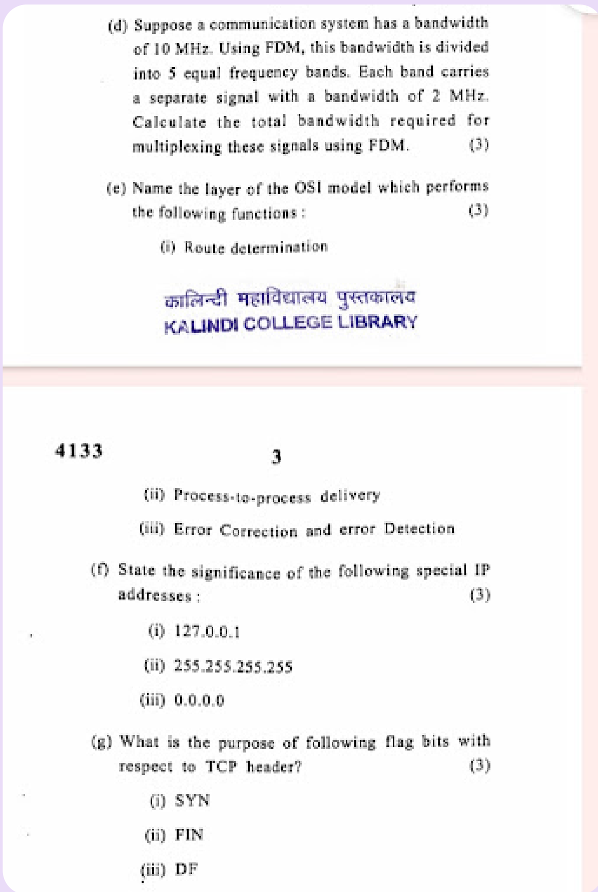
- Bandwidth (B) = $200 \text{ KHz} = 200,000 \text{ Hz}$
- Signal levels (L) = 16

Calculation: First, determine the number of bits per signal level: $\log_2(16) = 4 \text{ bits per level}$

Next, plug the values into the Nyquist formula: $C = 2 \times 200,000 \times 4$ $C = 400,000 \times 4$ $C = 1,600,000 \text{ bps}$

Answer: The maximum data rate of the channel is 1.6 Mbps (Megabits per second).

You Asked



(d) Suppose a communication system has a bandwidth of 10 MHz. Using FDM, this bandwidth is divided into 5 equal frequency bands. Each band carries a separate signal with a bandwidth of 2 MHz. Calculate the total bandwidth required for multiplexing these signals using FDM. (3)

(e) Name the layer of the OSI model which performs the following functions : (3)

(i) Route determination

(ii) Process-to-process delivery

(iii) Error Correction and error Detection

(f) State the significance of the following special IP addresses : (3)

(i) 127.0.0.1

(ii) 255.255.255.255

(iii) 0.0.0.0

(g) What is the purpose of following flag bits with respect to TCP header? (3)

(i) SYN

(ii) FIN

(iii) DF

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Here are the solutions to the remaining networking questions from the provided image:

1. (d) FDM Bandwidth Calculation

In Frequency Division Multiplexing (FDM), the total required bandwidth is the sum of the bandwidths of the individual signals, plus any guard bands used to prevent interference.

Since the problem does not specify any guard bands, we calculate the total bandwidth by simply multiplying the number of signals by the bandwidth of each signal:

- Number of signals (bands) = 5
- Bandwidth of each signal = 2 MHz

$$\text{Total Bandwidth} = 5 \times 2 \text{ MHz} = \mathbf{10} \text{ MHz}$$

(Note: In a real-world scenario, the required bandwidth would be slightly larger than 10 MHz to accommodate guard bands between the channels).

---

1. (e) OSI Model Layers

- **(i) Route determination: Network Layer** (Layer 3). It uses routing protocols to determine the best path for data packets to travel from the source to the destination network.
- **(ii) Process-to-process delivery: Transport Layer** (Layer 4). It uses port numbers to ensure that data is delivered to the correct application/process running on the destination host.

- **(iii) Error Correction and error Detection: Data Link Layer** (Layer 2). It frames data and adds a Frame Check Sequence (like a CRC) to detect and sometimes correct errors that occur over the physical medium. *(Note: The Transport layer also performs end-to-end error detection, but the Data Link layer is responsible for node-to-node error control).*

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1. (f) Significance of Special IP Addresses

- **(i) 127.0.0.1:** This is the **Loopback Address**. It is used by a host to send packets to itself for network testing and inter-process communication without the data ever leaving the local machine or going onto the physical network wire.
- **(ii) 255.255.255.255:** This is the **Limited Broadcast Address**. It is used to send a packet to every single host on the local physical network segment. Routers will not forward these packets to other networks.
- **(iii) 0.0.0.0:** This is the **Unspecified Address**. It typically means "this host on this network." It is often used as a source address during the DHCP discovery process when a device is booting up and doesn't yet have its own IP address assigned. It is also used in routing tables to represent the "Default Route."

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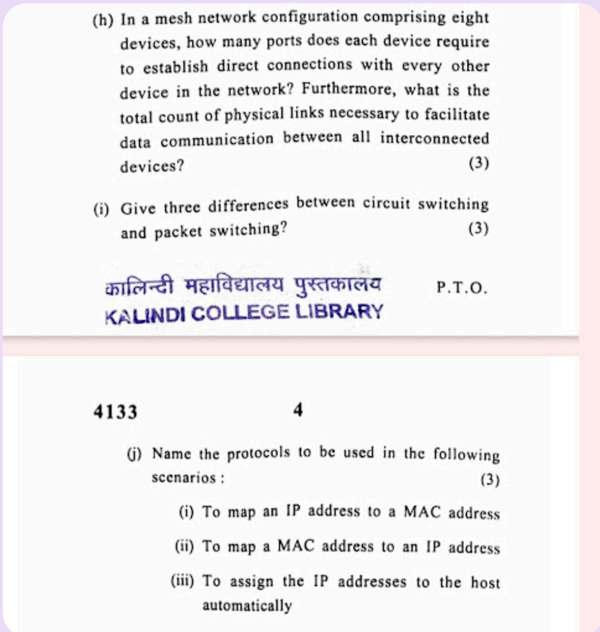
1. (g) TCP Header Flag Bits

(Note: There appears to be an error in the question paper regarding item (iii). DF is an IP header flag, not a TCP header flag. The answers below reflect their actual networking functions).

- **(i) SYN (Synchronize):** Used during the initial three-way handshake to establish a TCP connection. It synchronizes sequence numbers between the client and server so they can reliably track the data being sent.
- **(ii) FIN (Finish):** Used during connection termination. It indicates that the sender has no more data to transmit and wants to gracefully close the connection.
- **(iii) DF (Don't Fragment):** This flag is actually found in the **IPv4 Header**, not the TCP header. When set to 1, it instructs routers along the path *not* to fragment the packet if it exceeds the Maximum Transmission Unit (MTU) of the next link. If a router cannot forward the packet without fragmenting it, it will drop the packet and send an ICMP error message back to the sender.

You Asked



(h) In a mesh network configuration comprising eight devices, how many ports does each device require to establish direct connections with every other device in the network? Furthermore, what is the total count of physical links necessary to facilitate data communication between all interconnected devices? (3)

(i) Give three differences between circuit switching and packet switching? (3)

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(j) Name the protocols to be used in the following scenarios : (3)

(i) To map an IP address to a MAC address

(ii) To map a MAC address to an IP address

(iii) To assign the IP addresses to the host automatically

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Here are the solutions to the final set of networking questions from the provided image:

1. (h) Mesh Network Ports and Links Calculation

For a fully connected mesh network with n devices:

- **Ports per device:** Every device must have a dedicated point-to-point link to every *other* device in the network. Therefore, the formula is $n - 1.\, 8 - 1 = \mathbf{7}\ \text{ports per device}$
- **Total physical links:** Since every link connects two devices, the total number of unique physical cables needed is calculated using the formula $\frac{n(n-1)}{2}.\ \frac{8\times(8-1)}{2} = \frac{8\times 7}{2} = \mathbf{28}\ \text{physical links}$

---

1. (i) Differences between Circuit Switching and Packet Switching

Here are three key differences:

Feature	Circuit Switching	Packet Switching
Path Establishment	Requires a dedicated physical path to be established between source and destination before data transfer begins.	Does not establish a dedicated path; individual packets are dynamically routed independently.
Data Format	Data is transmitted as a continuous, steady stream.	Data is broken down into smaller, discrete blocks called packets before transmission.
Resource Allocation	Bandwidth is reserved for the entire duration of the connection, even if no data is currently being sent (leading to potential waste).	Bandwidth is shared dynamically among multiple users on demand, making it much more efficient.

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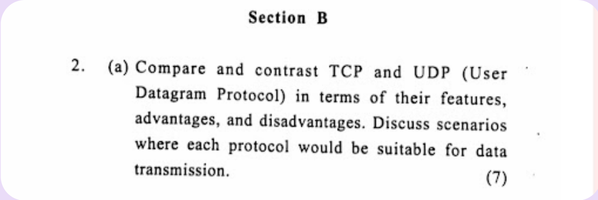
1. (j) Network Protocols

- **(i) To map an IP address to a MAC address: ARP** (Address Resolution Protocol). When a device knows the IP address but needs the physical MAC address to transmit over the local network, it broadcasts an ARP request.
- **(ii) To map a MAC address to an IP address: RARP** (Reverse Address Resolution Protocol). Historically used by diskless workstations to discover their own IP address based on their factory-burned MAC address. *(Note: Modern networks generally use DHCP to achieve this function, but RARP is the specific textbook protocol for this exact one-to-one mapping).*
- **(iii) To assign the IP addresses to the host automatically: DHCP** (Dynamic Host Configuration Protocol). It automatically assigns IP addresses, subnet masks, default gateways, and other network parameters to client devices upon joining a network.

You Asked


Section B

2. (a) Compare and contrast TCP and UDP (User Datagram Protocol) in terms of their features, advantages, and disadvantages. Discuss scenarios where each protocol would be suitable for data transmission. (7)

Gemini 3.1 Pro

Here is a comprehensive answer to the question presented in the image:

2. (a) Compare and Contrast TCP and UDP

Both **TCP (Transmission Control Protocol)** and **UDP (User Datagram Protocol)** are foundational Transport Layer (Layer 4) protocols in the TCP/IP suite, responsible for delivering data between applications on different devices. However, they operate in fundamentally different ways.

1. Feature Comparison

Feature	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Connection Type	Connection-oriented (requires a 3-way handshake before data transmission).	Connectionless (sends data immediately without prior setup).
Reliability	Highly reliable. Guarantees packet delivery.	Unreliable ("Best-effort" delivery). No guarantee packets will arrive.
Data Ordering	Sequenced. Reassembles packets in the exact order they were sent.	Unsequenced. Packets may arrive out of order.
Error Checking	Extensive error checking and recovery. Retransmits lost or corrupted packets.	Basic error checking (checksum). Discards corrupted packets without retransmitting.
Flow & Congestion Control	Yes. Adjusts transmission speed to prevent overwhelming the receiver or network.	No. Sends data at a constant rate regardless of network conditions.
Header Size	20 to 60 bytes (High overhead).	8 bytes (Low overhead).
Transmission Speed	Slower due to acknowledgments and connection management overhead.	Much faster due to the absence of acknowledgments and setup overhead.

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2. Advantages and Disadvantages

TCP

- **Advantages:**
  - **Guaranteed Delivery:** You can be certain that the data sent is the data received, entirely intact.
  - **In-Order Delivery:** Applications do not need to worry about sorting out-of-sequence packets.
  - **Network Friendly:** Flow and congestion control prevent network collapses during heavy traffic.
- **Disadvantages:**

- **High Latency/Slower:** The constant back-and-forth of acknowledgments and the initial setup take time.
- **Resource Intensive:** Requires more memory and processing power to track connection states and sequence numbers.

UDP

- **Advantages:**
  - **High Speed & Low Latency:** Data is fired off immediately, making it ideal for time-sensitive applications.
  - **Efficiency:** The tiny 8-byte header means more bandwidth is used for actual payload data rather than control information.
  - **Broadcast/Multicast Support:** Can easily send the same data to multiple recipients simultaneously (TCP is strictly one-to-one).
- **Disadvantages:**
  - **Data Loss:** Packets can be dropped silently. If a packet goes missing, the application must handle it (or ignore it).
  - **Unordered Delivery:** If packet order matters, the application layer must sort it out.

---

3. Suitable Scenarios for Data Transmission

Scenarios suitable for TCP: TCP is essential when **accuracy and completeness** are more important than real-time speed.

- **Web Browsing (HTTP/HTTPS):** Loading a webpage requires all text and images to load perfectly. A missing piece of HTML would break the page.
- **File Transfers (FTP/SFTP):** When downloading a document or software executable, dropping even a single byte corrupts the entire file.
- **Email (SMTP/IMAP):** Emails must arrive exactly as typed, with no missing words or corrupted attachments.

Scenarios suitable for UDP: UDP is preferred when **speed and real-time delivery** are critical, and occasional data loss is acceptable or unnoticeable.

- **Live Video/Audio Streaming & VoIP:** In a Skype call or Twitch stream, it is better to have a tiny momentary glitch (a dropped packet) than to freeze the entire video while waiting for a lost frame to be retransmitted.
- **Online Multiplayer Gaming:** Fast-paced games require immediate location and action updates. Old data (like where a player was 2 seconds ago) is useless, so retransmitting lost packets is counterproductive.
- **DNS Queries:** Converting a URL (like https://www.google.com/search?q=google.com) to an IP address requires a quick, single request-and-reply. Setting up a full TCP connection for such a small transaction would be inefficient.

You Asked

(b) Consider a network comprising five routers labeled as A, B, C, D, and E. The interconnections and their associated costs are given as follows :

- Link between A and B: Cost 2
- Link between A and C: Cost 4
- Link between B and C: Cost 1
- Link between B and D: Cost 5
- Link between C and D: Cost 3
- Link between C and E: Cost 7
- Link between D and E: Cost 2

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Using Dijkstra's algorithm, illustrate the step-by-step process of determining the shortest path from router A to router E. (8)

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Here is the step-by-step illustration of determining the shortest path from router A to router E using Dijkstra's algorithm.

Initial Setup

Let's list the interconnections and costs (weights) given in the problem:

- A-B = 2
- A-C = 4
- B-C = 1
- B-D = 5
- C-D = 3
- C-E = 7
- D-E = 2

We will track the shortest known distance to each node from A. We start by setting the distance to the starting node (A) to 0, and all other nodes to infinity (∞).

Initialization:

- Visited set = `{}`
- Distances: `A=0, B=∞, C=∞, D=∞, E=∞`

Step-by-Step Execution

Step 1: Evaluating from Node A

- **Current Node:** A (distance = 0)
- **Action:** Look at unvisited neighbors of A (which are B and C). Calculate their distance from A.
  - Distance to B via A = $0 + 2 = 2$. (Since $2 < \infty$, update B's distance to 2)
  - Distance to C via A = $0 + 4 = 4$. (Since $4 < \infty$, update C's distance to 4)

- **Mark A as visited.**
- *Current known shortest distances:* `A=0, B=2, C=4, D=∞, E=∞`
- *Visited:* `{A}`

Step 2: Evaluating from Node B

- **Current Node:** B (It has the smallest distance, 2, among unvisited nodes)
- **Action:** Look at unvisited neighbors of B (which are C and D).
  - Distance to C via B = distance(A to B) + cost(B to C) = $2 + 1 = 3$. (Since 3 is less than the current known distance of 4, update C's distance to 3. The new path to C is via B).
  - Distance to D via B = distance(A to B) + cost(B to D) = $2 + 5 = 7$. (Since $7 < \infty$, update D's distance to 7).
- **Mark B as visited.**
- *Current known shortest distances:* `A=0, B=2, C=3, D=7, E=∞`
- *Visited:* `{A, B}`

Step 3: Evaluating from Node C

- **Current Node:** C (It has the smallest distance, 3, among unvisited nodes)
- **Action:** Look at unvisited neighbors of C (which are D and E).
  - Distance to D via C = distance(A to C) + cost(C to D) = $3 + 3 = 6$. (Since 6 is less than the current known distance of 7, update D's distance to 6. The new path to D is via C).
  - Distance to E via C = distance(A to C) + cost(C to E) = $3 + 7 = 10$. (Since $10 < \infty$, update E's distance to 10).
- **Mark C as visited.**
- *Current known shortest distances:* `A=0, B=2, C=3, D=6, E=10`
- *Visited:* `{A, B, C}`

Step 4: Evaluating from Node D

- **Current Node:** D (It has the smallest distance, 6, among unvisited nodes)
- **Action:** Look at unvisited neighbors of D (which is E).
  - Distance to E via D = distance(A to D) + cost(D to E) = $6 + 2 = 8$. (Since 8 is less than the current known distance of 10, update E's distance to 8. The new path to E is via D).
- **Mark D as visited.**
- *Current known shortest distances:* `A=0, B=2, C=3, D=6, E=8`
- *Visited:* `{A, B, C, D}`

Step 5: Evaluating from Node E

- **Current Node:** E (It is the destination and the only remaining unvisited node)
- Since it has no unvisited neighbors, we mark E as visited and the algorithm concludes.
- *Visited:* `{A, B, C, D, E}`

Summary Table

Let $D(v)$ be the shortest distance to node v, and $p(v)$ be its predecessor node in the shortest path.

Step	Visited Nodes	$D(B), p(B)$	$D(C), p(C)$	$D(D), p(D)$	$D(E), p(E)$
0	{A}	$2, A$	$4, A$	∞	∞
1	{A, B}		$\mathbf{3, B}$	$7, B$	∞
2	{A, B, C}			$\mathbf{6, C}$	$10, C$
3	{A, B, C, D}				$\mathbf{8, D}$
4	{A, B, C, D, E}				

Final Result

By tracing the predecessors backward from E:

- E comes from D
- D comes from C
- C comes from B
- B comes from A

The shortest path from router A to router E is: **A** $\longrightarrow$ **B** $\longrightarrow$ **C** $\longrightarrow$ **D** $\longrightarrow$ **E**

The total minimum cost for this path is **8**.

You Asked

3. (a) Consider a Go-Back-N ARQ protocol operating over a network. The window size (W) is 4, and the sequence numbers range from 0 to 7. Assume that frames with sequence numbers 0, 1,2, and 3 have been successfully received by the receiver, and the receiver has sent acknowledgment (ACK) for frames up to sequence number 3. Now, due to network congestion, the frame with sequence number 4 is lost. (7)

   (i) What is the size of the window for sender and receiver?

   (ii) Calculate the range of valid sequence numbers at the sender's side after the loss of frame with sequence number 4.

   (iii) Determine the range of sequence numbers expected by the receiver after it has sent ACK for frames up to sequence number 3.

   (iv) Explain how the sender and receiver handle the loss of frame 4 and maintain synchronization using sequence numbers in the Go-Back-N ARQ protocol.

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Here are the solutions to the Go-Back-N ARQ protocol questions presented in the image:

3. (a) Go-Back-N ARQ Protocol

(i) What is the size of the window for sender and receiver?

In the Go-Back-N (GBN) ARQ protocol:

- **Sender Window Size (W_s):** The problem explicitly states the window size is 4. So, the sender window size is **4**.
- **Receiver Window Size (W_r):** In Go-Back-N, the receiver only accepts frames in strict, sequential order. Therefore, the receiver window size is always **1**.

(ii) Calculate the range of valid sequence numbers at the sender's side after the loss of frame with sequence number 4.

The sender has a window size of 4. Since frames 0, 1, 2, and 3 have been successfully received and acknowledged, the sender's window slides forward past these acknowledged frames.

- The lowest unacknowledged frame is now 4.
- With a window size of 4, the sender is allowed to transmit the next four consecutive frames.
- Therefore, the valid range of sequence numbers currently in the sender's window is **4, 5, 6, and 7** (or the range **[4, 7]**).

(iii) Determine the range of sequence numbers expected by the receiver after it has sent ACK for frames up to sequence number 3.

Because the receiver window size in Go-Back-N is strictly 1, it only ever expects one specific frame at a time.

- It has successfully received and acknowledged up to frame 3.
- Therefore, its window of size 1 slides forward to the very next sequence number.
- The receiver is exclusively expecting sequence number **4** (the range is just **[4]**).

(iv) Explain how the sender and receiver handle the loss of frame 4 and maintain synchronization using sequence numbers in the Go-Back-N ARQ protocol.

Here is the step-by-step process of how GBN handles this loss to maintain synchronization:

1. **Sender continues transmitting:** The sender transmits frame 4 (which gets lost), but because its window size is 4, it continues to transmit frames 5, 6, and 7 without waiting for an ACK for frame 4. It also starts a timer for the oldest unacknowledged frame (frame 4).
2. **Receiver discards out-of-order frames:** The receiver is strictly expecting sequence number 4. When it receives frames 5, 6, and 7, it recognizes they are out of order. In Go-Back-N, the receiver **discards** these out-of-order frames entirely. It typically replies to these out-of-order frames by sending a duplicate ACK for the last correctly received frame (ACK 3), signaling to the sender "I am still waiting for frame 4."
3. **Sender times out:** Eventually, the sender's timer for frame 4 expires because it never received a valid acknowledgment for it.
4. **Go-Back-N Retransmission:** Upon timeout, the sender "goes back N" steps. It resets its transmission state and **retransmits frame 4, as well as all subsequent frames that were in its window** (frames 5, 6, and 7), regardless of whether those later frames actually reached the receiver the first time.
5. **Synchronization restored:** The receiver gets the retransmitted frame 4, accepts it, and shifts its window to expect frame 5, allowing the sequential data transfer to resume smoothly.

You Asked

(b) Differentiate between the following : (8)

(i) Unipolar and Bipolar Line Coding Schemes

(ii) Data Element and Signal Element

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Here are the differentiations for the topics presented in the image:

3. (b) Differentiate between the following:

(i) Unipolar and Bipolar Line Coding Schemes

Line coding is the process of converting digital data (sequences of bits) into digital signals. Unipolar and Bipolar are two different categories of these schemes.

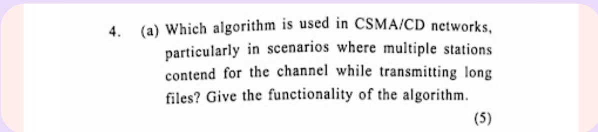
Feature	Unipolar Line Coding	Bipolar Line Coding
Voltage Levels	Uses only **two** voltage levels: one non-zero (usually positive) and zero.	Uses **three** voltage levels: positive (+V), negative (-V), and zero (0).
Representation (Typical)	Bit `1` is represented by a positive voltage (+V). Bit `0` is represented by zero voltage (0V).	Bit `0` is represented by zero voltage. Bit `1` is represented by alternating positive (+V) and negative (-V) voltages.
DC Component	Has a significant DC component (average voltage is non-zero). This can cause issues in systems that cannot pass low frequencies.	The DC component is completely eliminated because the alternating positive and negative pulses for `1`s cancel each other out over time.
Synchronization	Poor synchronization. A long string of `1`s or `0`s results in a flat signal, making it hard for the receiver to clock the individual bits.	Better synchronization, especially for strings of `1`s, since the signal transitions between positive and negative. (Though long strings of `0`s can still be an issue without modifications like B8ZS or HDB3).
Common Examples	Unipolar NRZ (Non-Return-to-Zero).	AMI (Alternate Mark Inversion), Pseudoternary.
Application/Cost	Simple to implement and cheap, but rarely used in long-distance modern networking due to its flaws.	More complex but widely used in long-distance communication (like T1 lines) due to better signal properties.

(ii) Data Element and Signal Element

In data communications, it is crucial to distinguish between the information being sent and the physical signal carrying it.

Feature	Data Element	Signal Element
Definition	The smallest piece of information to be transmitted.	The shortest meaningful unit of a digital signal being transmitted over a medium.
Nature	It is the abstract information itself (what we need to send).	It is the physical vehicle or carrier (how we send it).
Basic Unit	The basic unit is the **Bit** (Binary Digit: 0 or 1).	The basic unit is a pulse or a specific signal state (e.g., a specific voltage level, phase, or frequency).
Rate Measurement	Measured as **Data Rate** or **Bit Rate** (bits per second - bps).	Measured as **Signal Rate**, **Pulse Rate**, or **Baud Rate** (baud).
Relationship	A single data element can be carried by one or multiple signal elements.	A single signal element can carry one data element, multiple data elements, or even a fraction of a data element, depending on the encoding scheme.
Analogy	Think of data elements as the **passengers** needing transport.	Think of signal elements as the **vehicles** (cars, buses) carrying those passengers.

You Asked


4. (a) Which algorithm is used in CSMA/CD networks, particularly in scenarios where multiple stations contend for the channel while transmitting long files? Give the functionality of the algorithm. (5)

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Here is the solution to the networking question presented in the image:

4. (a) Algorithm used in CSMA/CD Networks

Name of the Algorithm: The algorithm used in CSMA/CD (Carrier Sense Multiple Access with Collision Detection) networks to manage collisions and channel contention is the **Truncated Binary Exponential Backoff (BEB) Algorithm**.

Functionality of the Algorithm: When multiple stations attempt to transmit simultaneously, a collision occurs. The BEB algorithm dictates how long each station should wait before attempting to retransmit, dynamically adjusting the wait time based on network traffic load.

Here is the step-by-step functionality:

1. **Collision Detection:** If a station detects a collision while transmitting, it immediately stops sending data and broadcasts a short "jam signal" to ensure all other stations know a collision occurred.
2. **Attempt Counter:** The station maintains a collision counter, let's call it c, which tracks the number of consecutive collisions for the current frame. Initially, $c = 0$. After a collision, c is incremented ($c = c + 1$).

3. **Maximum Attempt Limit:** If the collision counter reaches a predefined maximum limit (typically 16 in Ethernet), the station aborts the transmission entirely and reports a failure to the upper network layers.
4. **Calculating the Contention Window:** If $c \le 16$, the station calculates a "contention window" (a range of numbers). The range is from 0 to $2^k - 1$, where k is the number of collisions, but capped at a maximum value (usually 10).
    - Mathematically: $k = \min(c, 10)$
    - The window size grows exponentially:
        - 1st collision: range is 0 to 1 ($2^1 - 1$)
        - 2nd collision: range is 0 to 3 ($2^2 - 1$)
        - 3rd collision: range is 0 to 7 ($2^3 - 1$)
        - ...
        - 10th collision and beyond: range is 0 to 1023 ($2^{10} - 1$)
5. **Random Wait Time:** The station picks a random integer number, r, from this calculated contention window.
6. **Backoff Delay:** The station waits for a duration equal to $r \times \text{slot\_time}$ (where slot time is a fixed network parameter, typically 51.2 microseconds for standard 10 Mbps Ethernet).
7. **Retransmission Attempt:** After the backoff timer expires, the station goes back to the basic CSMA rules: it senses the carrier. If the channel is idle, it begins transmission again. If the channel is busy, it waits until it becomes idle.

Why this is effective for long files: When transmitting long files, stations are continuously trying to seize the channel, leading to high contention. If they used a fixed wait time, they would likely collide repeatedly. The exponential backoff spreads out the retransmission attempts over an increasingly wider timeframe. If collisions keep happening, the algorithm assumes the network is heavily congested and forces stations to wait longer on average, effectively smoothing out the traffic and preventing a complete network collapse.

You Asked

(b) Given the IP address 192.168.0.0/24, an organization needs to create two subnets with the following requirements: (10)

- Subnet 1: 50 hosts
- Subnet 2: 30 hosts

For each of these subnets, give the

(i) first IP address assigned

(ii) last IP address assigned

(iii) subnet mask in the w.x.y.z/s notation

(iv) identify the range of IP addresses available

(v) number of host and network bits

Gemini 3.1 Pro

To allocate the IP addresses efficiently based on the specific host requirements, we must use **Variable Length Subnet Masking (VLSM)**. This technique allows us to allocate subnet masks of different sizes to different subnets within the same base network, thereby minimizing the waste of unassigned IP addresses.

We are given the base network block **192.168.0.0/24**, which provides a total of 8 host bits (since $32 - 24 = 8$). This equates to $2^8 = 256$ total addresses.

When using VLSM, the standard engineering practice is to always satisfy the requirement for the largest subnet first, and then proceed in descending order.

Subnet 1: Requirement of 50 Hosts

1. Calculating the Required Bits To find the necessary number of host bits (h), we use the formula for usable hosts: $2^h - 2 \geq \text{Required Hosts}$ We subtract 2 because the first address of any subnet is reserved for the Network ID, and the last address is reserved for the Broadcast ID.

For 50 hosts: $2^h - 2 \geq 50$ The smallest integer h that satisfies this inequality is **6**. ($2^6 - 2 = 64 - 2 = 62$ usable hosts, which comfortably accommodates the required 50).

2. Calculating the Network Bits and Subnet Mask An IPv4 address consists of 32 bits. Since we are reserving 6 bits for the hosts, the network bits will be: $32 - 6 = 26 \text{ network bits}$ This gives us a CIDR notation of **/26**. To find the dotted-decimal subnet mask, we convert the 26 network bits (represented as `1` s) and 6 host bits (represented as `0` s) into decimal format: `11111111.11111111.11111111.11000000` $\rightarrow$ **255.255.255.192**

3. Defining the Subnet Boundaries

- The block size (total addresses) is $2^6 = 64$.
- Since this is the first subnet, it starts at the very beginning of our given block: **192.168.0.0**.
- The network ID is `192.168.0.0`.
- The broadcast ID is one less than the start of the next block: `192.168.0.63`.

Subnet 1 Final Answers:

- **(i) First IP address assigned:** 192.168.0.1
- **(ii) Last IP address assigned:** 192.168.0.62
- **(iii) Subnet mask in w.x.y.z/s notation:** 255.255.255.192 /26
- **(iv) Identify the range of IP addresses available:** 192.168.0.1 – 192.168.0.62
- **(v) Number of host and network bits:** Host bits: 6, Network bits: 26

Subnet 2: Requirement of 30 Hosts

1. Calculating the Required Bits Using the same formula for the remaining requirement of 30 hosts: $2^h - 2 \geq 30$ The smallest integer h that satisfies this inequality is **5**. ($2^5 - 2 = 32 - 2 = 30$ usable hosts, which is a perfect fit for the requirement).

2. Calculating the Network Bits and Subnet Mask Subtracting the 5 host bits from the total 32 bits of an IPv4 address: $32 - 5 = 27 \text{ network bits}$ This gives us a CIDR notation of **/27**. To find the dotted-decimal subnet mask, we convert the 27 network bits and 5 host bits into decimal format: `11111111.11111111.11111111.11100000` $\rightarrow$ **255.255.255.224**

3. Defining the Subnet Boundaries

- The block size (total addresses) is $2^5 = 32$.
- This subnet begins exactly where Subnet 1 ended. Subnet 1's broadcast was `.63`, so Subnet 2 starts at **192.168.0.64**.
- The network ID is `192.168.0.64`.

- Adding the block size of 32 means the next theoretical subnet would start at `.96` (`64 + 32`). Therefore, the broadcast ID for Subnet 2 is `192.168.0.95`.

Subnet 2 Final Answers:

- **(i) First IP address assigned:** 192.168.0.65
- **(ii) Last IP address assigned:** 192.168.0.94
- **(iii) Subnet mask in w.x.y.z/s notation:** 255.255.255.224 /27
- **(iv) Identify the range of IP addresses available:** 192.168.0.65 – 192.168.0.94
- **(v) Number of host and network bits:** Host bits: 5, Network bits: 27

You Asked

5. (a) Assume that a voice channel occupies a bandwidth of 4 kHz. We need to multiplex 10 voice channels with guard bands of 500 Hz using FDM. Calculate the required bandwidth. (5)

Gemini 3.1 Pro

Here is the step-by-step solution to the Frequency Division Multiplexing (FDM) calculation problem:

5. (a) FDM Required Bandwidth Calculation

1. Identify the Given Values:

- Number of voice channels (N) = 10
- Bandwidth of each voice channel (B_c) = 4 kHz
- Bandwidth of each guard band (B_g) = $500 \text{ Hz} = 0.5 \text{ kHz}$

2. Understand the Logic of FDM: In Frequency Division Multiplexing, multiple signals are combined into one medium by assigning each signal a different frequency range. To prevent these signals from overlapping and causing interference, empty frequency spaces called **guard bands** are placed *between* adjacent channels.

If you have N channels stacked side-by-side, you only need guard bands in the spaces between them. Therefore, the number of guard bands required is $(N - 1)$.

3. Perform the Calculations:

- **Total bandwidth needed for the channels:** $N \times B_c = 10 \times 4 \text{ kHz} = 40 \text{ kHz}$
- **Total number of guard bands:** $N - 1 = 10 - 1 = 9 \text{ guard bands}$
- **Total bandwidth needed for the guard bands:** $9 \times 0.5 \text{ kHz} = 4.5 \text{ kHz}$
- **Total Required Bandwidth:** $\text{Total Bandwidth} = \text{Total Channel Bandwidth} + \text{Total Guard Band Bandwidth}$ $\text{Total Bandwidth} = 40 \text{ kHz} + 4.5 \text{ kHz} = \mathbf{44.5} \text{ kHz}$

Answer: The total required bandwidth to multiplex the 10 voice channels with the specified guard bands is 44.5 kHz.

You Asked

(b) (i) A code includes the four four-bit codewords: 1001, 0110, 1010, and 0101.

- What is the minimum distance of this code?
- What is the maximum number of errors that this code is guaranteed to detect?
- What is the maximum number of errors that this code is guaranteed to correct?

(ii) A 12-bit odd-parity Hamming code whose binary value is 111001001111 arrives at a receiver. What was the original value of the message ? Assume that not more than 1 bit is in error. (4+6)

Gemini 3.1 Pro

Here are the step-by-step solutions to the error-correction coding questions in the image:

(b) (i) Analysis of the 4-bit Code

The given code consists of four codewords: `1001`, `0110`, `1010`, and `0101`.

1. What is the minimum distance of this code? To find the minimum distance (d_{min}), we calculate the Hamming distance between every possible pair of codewords. The Hamming distance is the number of bit positions where two codewords differ.

- $d(1001, 0110) = 4$
- $d(1001, 1010) = 2$
- $d(1001, 0101) = 2$
- $d(0110, 1010) = 2$
- $d(0110, 0101) = 2$
- $d(1010, 0101) = 4$

The smallest value among these is 2. **Answer:** The minimum distance of this code is **2**.

2. What is the maximum number of errors that this code is guaranteed to detect? The formula for the maximum number of detectable errors (s) is: $s = d_{min} - 1$ $s = 2 - 1 = 1$ **Answer:** This code is guaranteed to detect up to **1 error**.

3. What is the maximum number of errors that this code is guaranteed to correct? The formula for the maximum number of correctable errors (t) is: $t = \lfloor (d_{min} - 1)/2 \rfloor$ $t = \lfloor (2 - 1)/2 \rfloor = \lfloor 0.5 \rfloor = 0$ **Answer:** This code cannot reliably correct any errors (**0 errors**). If one bit flips, it results in a sequence that is exactly 1 bit away from two different valid codewords, making it impossible to know which was the original.

(b) (ii) 12-bit Odd-Parity Hamming Code

We received the 12-bit binary sequence: `111001001111`. A 12-bit Hamming code consists of 4 parity bits (P_1, P_2, P_4, P_8) located at positions that are powers of 2 (1, 2, 4, 8) and 8 data bits located at the remaining positions (3, 5, 6, 7, 9, 10, 11, 12).

Important Note on Indexing: If we apply standard Left-to-Right indexing (where the leftmost bit is position 1), the resulting error syndrome is 13, which is outside the 12-bit boundary and contradicts the "not more than 1 bit in error" constraint. Therefore, standard digital logic **Right-to-Left indexing** (where the LSB/rightmost bit is position 1) must be used for this sequence.

Let's map the received string `111001001111` right-to-left:

Bit Position	12	11	10	9	8	7	6	5	4	3	2	1
Role	D_{12}	D_{11}	D_{10}	D_9	P_8	D_7	D_6	D_5	P_4	D_3	P_2	P_1
Value	1	1	1	0	**0**	1	0	0	**1**	1	**1**	**1**

Now, we calculate the error syndrome checks (C_1, C_2, C_4, C_8). Because this is an **odd-parity** system, every checked group must contain an *odd* number of '1's.

- If the sum of '1's is **odd**, the parity check passes (Syndrome bit = **0**).
- If the sum of '1's is **even**, there is an error (Syndrome bit = **1**).

Step 1: Perform the Parity Checks

- C_1 **checks positions 1, 3, 5, 7, 9, 11:** Bits are `1, 1, 0, 1, 0, 1`. The total number of '1's is 4 (Even). Result: Error $\rightarrow \mathbf{C_1 = 1}$
- C_2 **checks positions 2, 3, 6, 7, 10, 11:** Bits are `1, 1, 0, 1, 1, 1`. The total number of '1's is 5 (Odd). Result: Pass $\rightarrow \mathbf{C_2 = 0}$
- C_4 **checks positions 4, 5, 6, 7, 12:** Bits are `1, 0, 0, 1, 1`. The total number of '1's is 3 (Odd). Result: Pass $\rightarrow \mathbf{C_4 = 0}$
- C_8 **checks positions 8, 9, 10, 11, 12:** Bits are `0, 0, 1, 1, 1`. The total number of '1's is 3 (Odd). Result: Pass $\rightarrow \mathbf{C_8 = 0}$

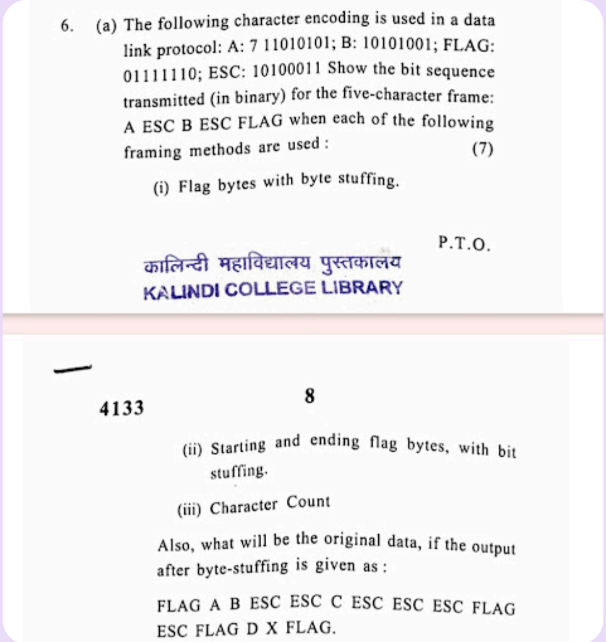
Step 2: Determine the Error Position The syndrome word is formed by combining the check results: $C_8C_4C_2C_1$. Syndrome = `0001` (Binary) = **1** (Decimal). This means the error occurred at **bit position 1**.

Step 3: Correct the Error The bit at position 1 was received as a `1`. To correct it, we flip it to a `0`. The corrected 12-bit codeword is: `11100100111` **`0`**

Step 4: Extract the Original Message To find the original message, we pull out just the data bits ($D_{12}, D_{11}, D_{10}, D_9, D_7, D_6, D_5, D_3$) from the corrected codeword: Message = `1 1 1 0 1 0 0 1`

Answer: The original value of the message was **`11101001`**.

You Asked



6. (a) The following character encoding is used in a data link protocol: A: 7 11010101; B: 10101001; FLAG: 01111110; ESC: 10100011 Show the bit sequence transmitted (in binary) for the five-character frame: A ESC B ESC FLAG when each of the following framing methods are used : (7)

(i) Flag bytes with byte stuffing.

P.T.O.

4133 8

(ii) Starting and ending flag bytes, with bit stuffing.

(iii) Character Count

Also, what will be the original data, if the output after byte-stuffing is given as :

FLAG A B ESC ESC C ESC ESC ESC FLAG ESC FLAG D X FLAG.

Gemini 3.1 Pro

Based on the character encodings provided in the image, let's break down the solutions for each framing method.

Given Encodings (Binary):

- **A:** `11010101` *(assuming the artifact before the '1' is a typo)*
- **B:** `10101001`
- **FLAG:** `01111110`
- **ESC:** `10100011`

Original Payload to Transmit: `A ESC B ESC FLAG`

(i) Flag bytes with byte stuffing

Rule: Add a `FLAG` byte at the beginning and end of the frame. To prevent the receiver from accidentally interpreting payload data as a control character, insert an `ESC` byte immediately before any `ESC` or `FLAG` byte that appears naturally within the payload.

1. **Analyze Payload:** `A`, `ESC`, `B`, `ESC`, `FLAG`
2. **Apply Stuffing:**
   - `A` remains `A`
   - `ESC` becomes `ESC` `ESC`
   - `B` remains `B`
   - `ESC` becomes `ESC` `ESC`
   - `FLAG` becomes `ESC` `FLAG`
3. **Stuffed Payload:** `A ESC ESC B ESC ESC ESC FLAG`
4. **Add Start/End Flags:** `FLAG A ESC ESC B ESC ESC ESC FLAG FLAG`

Transmitted Bit Sequence:
`01111110 11010101 10100011 10100011 10101001 10100011 10100011 10100011 01111110 01111110`

(ii) Starting and ending flag bytes, with bit stuffing

Rule: The frame begins and ends with the `01111110` flag. The payload is treated as a continuous stream of bits. To ensure the flag pattern never accidentally appears in the data, a `0` is stuffed (inserted) immediately after any five consecutive `1` s in the payload.

1. **Original Payload (Binary):** `11010101 10100011 10101001 10100011 01111110`
2. **Locate Consecutive 1s:** Scanning the stream, the only place with five consecutive `1` s is within the final `FLAG` character of the payload: `...01111110`
3. **Apply Bit Stuffing:** Insert a `0` after the fifth `1` : `...011111` **`0`** `10`
4. **Stuffed Payload:** `11010101 10100011 10101001 10100011 011111010`
5. **Add Start/End Flags:**

Transmitted Bit Sequence: `01111110 11010101 10100011 10101001 10100011 011111010 01111110`

(iii) Character Count

Rule: The first byte of the frame contains the total character count for the entire frame (including the count byte itself). No stuffing is used.

1. **Determine Count:** The payload has 5 characters (`A` , `ESC` , `B` , `ESC` , `FLAG`). Adding the count byte itself means the total frame length is **6** characters.
2. **Binary for 6:** `00000110`
3. **Frame Structure:** `[COUNT=6] A ESC B ESC FLAG`

Transmitted Bit Sequence: `00000110 11010101 10100011 10101001 10100011 01111110`

Reverse Byte-Stuffing (De-stuffing)

Given Output Frame: `FLAG A B ESC ESC C ESC ESC ESC FLAG ESC FLAG D X FLAG`

Rule for Receiver:

- The first and last `FLAG` s represent the frame boundaries and are discarded.
- When an `ESC` byte is encountered in the payload, the receiver discards it and treats the *immediately following byte* strictly as data, even if it is a control character.

De-stuffing Process:

1. `FLAG` → (Start Frame)
2. `A` → Data: **A**
3. `B` → Data: **B**
4. `ESC` `ESC` → Discard first ESC, keep second as data: **ESC**
5. `C` → Data: **C**
6. `ESC` `ESC` → Discard first ESC, keep second as data: **ESC**
7. `ESC` `FLAG` → Discard first ESC, keep FLAG as data: **FLAG**
8. `ESC` `FLAG` → Discard first ESC, keep FLAG as data: **FLAG**

9. D $\longrightarrow$ Data: **D**
10. X $\longrightarrow$ Data: **X**
11. FLAG $\longrightarrow$ (End Frame)

Original Data Payload: A B ESC C ESC FLAG FLAG D X

You Asked

7. (a) State Shannon's theorem. Determine the maximum achievable data rate for this channel based on Shannon's theorem when the communication channel has a bandwidth of 5 MHz and an SNR of 40 dB. (7)

Gemini 3.1 Pro

Here is the step-by-step solution to the problem presented in the image:

7. (a) Shannon's Theorem and Maximum Data Rate Calculation

Part 1: Statement of Shannon's Theorem

Shannon's Theorem (often referred to as the Shannon-Hartley theorem) defines the theoretical maximum rate at which error-free data can be transmitted over a communication channel that is subject to noise.

It states that the maximum achievable data rate (or channel capacity) is determined by the bandwidth of the channel and the ratio of the signal power to the noise power (Signal-to-Noise Ratio, or SNR).

The mathematical formula is: $C = B \times \log_2(1 + \text{SNR})$

Where:

- C = Channel Capacity (maximum achievable data rate) in bits per second (bps)
- B = Bandwidth of the channel in Hertz (Hz)
- SNR = Signal-to-Noise Ratio (this must be a linear ratio, not in decibels)

Part 2: Calculation

Given Values:

- Bandwidth (B) = $5 \text{ MHz} = 5,000,000 \text{ Hz}$ (or $5 \times 10^6 \text{ Hz}$)
- Signal-to-Noise Ratio (SNR_{dB}) = 40 dB

Step 1: Convert SNR from decibels (dB) to a linear ratio. The formula given in the problem is in decibels, but Shannon's equation requires the linear power ratio. We convert it using the standard formula: $SNR_{dB} = 10 \times \log_{10}(\text{SNR})$ $40 = 10 \times \log_{10}(\text{SNR})$ $4 = \log_{10}(\text{SNR})$ $\text{SNR} = 10^4 = 10,000$

Step 2: Apply the values to Shannon's formula. $C = B \times \log_2(1 + \text{SNR})$ $C = (5 \times 10^6) \times \log_2(1 + 10,000)$ $C = (5 \times 10^6) \times \log_2(10,001)$

Step 3: Calculate the logarithm and solve. *(Note:* $\log_2(x) = \frac{\log_{10}(x)}{\log_{10}(2)}$) $\log_2(10,001) \approx 13.2878$

$C \approx 5,000,000 \times 13.2878$ $C \approx 66,439,000 \text{ bps}$

Final Answer: The maximum achievable data rate for this channel is approximately 66.44 Mbps (Megabits per second).

You Asked

(b) Draw the layered architecture of TCP/IP model explaining the services provided by each layer. Given an example network, if we change the LAN technology to a new one, which layers in the TCP/IP protocol suite need to be changed and why? List the layers at which routers and switches are used. (8)

Gemini 3.1 Pro

Here are the solutions to the final networking question presented in the image:

7. (b) TCP/IP Model Architecture and Hardware

1. Layered Architecture of the TCP/IP Model and Services

The standard TCP/IP model consists of four distinct layers. Here is the architecture from top to bottom, along with the services provided by each:

- **Application Layer (Layer 4):**
  - **Services:** This is the highest layer that directly interfaces with software applications. It provides the protocols and services required for users to access the network. It handles functions like file transfer, email, web browsing, and domain name resolution.
  - **Common Protocols:** HTTP, HTTPS, FTP, SMTP, DNS, SSH.
- **Transport Layer (Layer 3):**
  - **Services:** This layer is responsible for end-to-end communication and data delivery between applications on different host computers. It uses port numbers to multiplex data to the correct application. It handles data segmentation, connection multiplexing, and can provide reliable data transfer (error recovery, flow control) or fast, best-effort transfer.
  - **Common Protocols:** TCP (Transmission Control Protocol), UDP (User Datagram Protocol).
- **Internet Layer (Layer 2):**
  - **Services:** Also known as the Network layer, its primary responsibility is routing packets across multiple independent networks to their final destination. It handles logical addressing, packet fragmentation, and path determination.
  - **Common Protocols:** IP (IPv4, IPv6), ICMP, ARP (sometimes placed between Network Access and Internet).
- **Network Access Layer (Layer 1):**

- **Services:** Also called the Link Layer or Network Interface Layer, it is the lowest layer. It encompasses the functions of both the OSI Data Link and Physical layers. It handles the physical transmission of data bits over the network medium (cables, wireless) and manages hardware addressing (MAC addresses), framing, and access control to the physical medium.
- **Common Protocols/Technologies:** Ethernet, Wi-Fi (802.11), PPP, Fiber Distributed Data Interface (FDDI).

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2. Changing the LAN Technology

- **Which layers need to be changed?** Only the **Network Access Layer** (the lowest layer).
- **Why?** The TCP/IP model is designed with the principle of layer independence and hardware abstraction. The upper layers (Internet, Transport, and Application) are completely decoupled from the underlying physical hardware. If you change your LAN from Ethernet to Wi-Fi, you only need to change the physical Network Interface Card (NIC) and the drivers associated with the Network Access Layer. The IP address (Internet layer) and TCP/UDP ports (Transport layer) remain exactly the same because IP does not care what type of wire or radio wave is carrying its packets.

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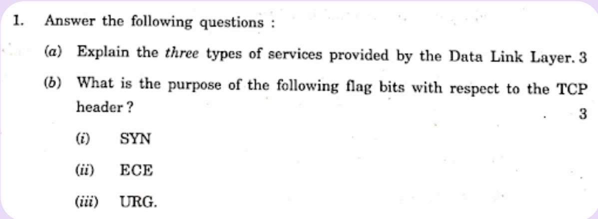
3. Layers for Routers and Switches

- **Routers:** Operate at the **Internet Layer** (Layer 2 of TCP/IP, or Layer 3 of the OSI model). They use logical IP addresses to determine the best path to forward packets across different networks.
- **Switches:** Standard network switches operate at the **Network Access Layer** (specifically the Data Link sub-layer, Layer 2 of the OSI model). They use physical MAC addresses to forward frames only to the specific device on a local network segment. *(Note: Multi-layer switches exist that can also perform routing, but a standard switch is strictly a Network Access/Layer 2 device).*

You Asked



1. Answer the following questions :

(a) Explain the *three* types of services provided by the Data Link Layer. 3

(b) What is the purpose of the following flag bits with respect to the TCP header? 3

(i) SYN

(ii) ECE

(iii) URG.

Gemini 3.1 Pro

Here are the answers to the questions presented in the image:

1. (a) Three types of services provided by the Data Link Layer

The Data Link Layer typically provides three main types of services to the Network Layer:

1. **Unacknowledged Connectionless Service:**
    - The source machine sends independent frames to the destination machine without having the destination acknowledge them.
    - No logical connection is established before sending, and none is released afterward.

- If a frame is lost due to noise, no attempt is made to recover it at this layer.
- **Use case:** Appropriate when the error rate is very low (so recovery is left to higher layers) or for real-time traffic (like voice or video) where late data is worse than bad data.

2. **Acknowledged Connectionless Service:**
   - There are no logical connections established, but each frame sent is individually acknowledged by the receiver.
   - The sender knows whether a frame has arrived correctly. If it doesn't arrive within a specific timeframe, it can be retransmitted.
   - **Use case:** Useful over unreliable channels, such as wireless networks (like 802.11 Wi-Fi), where frame loss is more common.
3. **Acknowledged Connection-Oriented Service:**
   - This is the most sophisticated service. The source and destination establish a logical connection before any data is transferred.
   - Each frame sent over the connection is numbered, and the Data Link Layer guarantees that every frame is received, exactly once, and in the correct order.
   - It involves three distinct phases: connection establishment, data transfer, and connection release.
   - **Use case:** Used for reliable, long-distance communication where higher layers expect a guaranteed, ordered data stream.

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1. (b) Purpose of the TCP header flag bits

- **(i) SYN (Synchronize):**
  - This flag is used during the initial setup of a TCP connection (the three-way handshake).
  - Its primary purpose is to synchronize sequence numbers between the client and server so both sides can keep track of the data stream being sent and received.
- **(ii) ECE (ECN-Echo):**
  - ECE stands for Explicit Congestion Notification Echo. It is used to handle network congestion.
  - It has two roles: If the SYN flag is also set to 1, it tells the other side that this device is capable of participating in ECN. If the SYN flag is 0, it is used during normal data transmission to echo back to the sender that it received a packet marked with "Congestion Experienced" from a router along the path, prompting the sender to slow down its transmission rate.
- **(iii) URG (Urgent):**
  - When this flag is set to 1, it indicates that the "Urgent Pointer" field in the TCP header is valid and significant.
  - It tells the receiving machine that a portion of the incoming data is urgent and should be prioritized and processed immediately by the application, bypassing the normal queue of incoming data.

You Asked

(c) State the purpose of the following IP addresses : 3

(i) 0.0.0.0

(ii) 127.xx.yy.zz

(iii) 255.255.255.255

(d) Name the layer of the TCP/IP model which performs the following services : 3

(i) Process-to-Process Delivery

(ii) Source-to-Destination Delivery

(iii) Framing.

Gemini 3.1 Pro

Here are the answers to the questions presented in the image:

(c) Purpose of the following IP addresses

(i) 0.0.0.0 This is the **Unspecified Address**. It generally means "this host on this network." Its most common purpose is serving as the source IP address during the initial phase of the DHCP process, when a device is booting up and requesting an IP address but hasn't been assigned one yet. It is also used in routing tables to designate the "Default Route" (the path to take when no specific route matches).

(ii) 127.xx.yy.zz This entire block (127.0.0.0 to 127.255.255.255) is reserved for **Loopback Addresses**. These addresses are used by a computer to send network traffic back to itself. This is primarily used for testing the local network stack, troubleshooting, and allowing software running on the same machine to communicate with each other via network protocols without the data ever reaching the physical network interface. (127.0.0.1 is the most commonly used).

(iii) 255.255.255.255 This is the **Limited Broadcast Address**. It is used when a device wants to send a single packet to every other device simultaneously on its local physical network segment. By design, routers will not forward packets addressed to 255.255.255.255, ensuring that the broadcast traffic stays confined to the local subnet.

(d) Layer of the TCP/IP model performing the following services

(i) Process-to-Process Delivery Transport Layer *(Explanation: The Transport layer uses port numbers to ensure that incoming data is delivered to the correct specific application or process running on the destination device.)*

(ii) Source-to-Destination Delivery Internet Layer (Also commonly referred to as the Network Layer) *(Explanation: The Internet layer uses logical IP addresses and routing protocols to determine the best path to move packets from the original source machine to the final destination machine across multiple different networks.)*

(iii) Framing Network Access Layer (Also commonly referred to as the Link Layer or Data Link Layer) *(Explanation: This layer is responsible for taking the packets from the Internet layer and encapsulating them into "frames." Framing involves adding a header (with physical MAC addresses) and a trailer (for error detection) so the data can be successfully transmitted over the physical medium.)*

You Asked

(e) Write the different kinds of transfer modes used in HDLC protocol. What is the purpose of using I-frame and S-frame ? 3

(f) State Shannon Capacity theorem. Consider an extremely noisy channel in which the signal-to-noise ratio is 40 dB. Compute the bit rate if the bandwidth of the channel is 1 kHz. 3

Gemini 3.1 Pro

Here are the solutions to the networking questions presented in the image:

(e) HDLC Protocol Transfer Modes and Frames

1. Transfer Modes in HDLC: The High-Level Data Link Control (HDLC) protocol utilizes three data transfer modes to govern communication between devices (typically a primary and one or more secondary stations):

- **Normal Response Mode (NRM):** An unbalanced configuration where the primary station initiates all communication and controls the link. Secondary stations can only transmit data in response to a direct command from the primary station.
- **Asynchronous Response Mode (ARM):** An unbalanced configuration similar to NRM, but with a key difference: secondary stations are allowed to transmit data frames without needing explicit permission from the primary station. The primary still retains overall control of the line (initialization, error recovery, etc.).
- **Asynchronous Balanced Mode (ABM):** A balanced configuration where all stations are treated as equals (combined stations). Any station can initiate a transmission without receiving permission from any other station. This is the most widely used mode today.

2. Purpose of I-frame and S-frame: HDLC defines three types of frames: Information (I), Supervisory (S), and Unnumbered (U).

- **I-frame (Information Frame):** The primary purpose of an I-frame is to transport the actual user data (payload) across the network. Additionally, I-frames can carry control information; they can piggyback acknowledgments (ACKs) for received data onto outgoing data frames, improving efficiency.
- **S-frame (Supervisory Frame):** S-frames do not carry user data. Their purpose is purely for link control, specifically for flow control and error control. They are used to send acknowledgments (ACKs) or negative acknowledgments (NAKs) when a station has no data to send (meaning piggybacking via an I-frame isn't possible).

(f) Shannon Capacity Theorem Calculation

1. State Shannon Capacity Theorem: The Shannon Capacity theorem (or Shannon-Hartley theorem) defines the theoretical maximum data transmission rate (capacity) of a communication channel that is subject to random noise. It states that the channel capacity is directly proportional to its bandwidth and the logarithm of the signal-to-noise ratio.

The mathematical formula is: $C = B \times \log_2(1 + \mathrm{SNR})$

Where:

- C = Channel Capacity (maximum bit rate) in bits per second (bps)
- B = Bandwidth of the channel in Hertz (Hz)
- SNR = Signal-to-Noise Ratio (as a linear power ratio, not in dB)

2. Computation:

- **Given values:**

- Signal-to-Noise Ratio (SNR_{dB}) = 40 dB
- Bandwidth (B) = 1 kHz = 1000 Hz

- **Step 1: Convert SNR from decibels (dB) to a linear ratio.** $SNR_{dB} = 10 \times \log_{10}(\text{SNR})$ $40 = 10 \times \log_{10}(\text{SNR})$ $4 = \log_{10}(\text{SNR})$ $\text{SNR} = 10^4 = 10,000$
- **Step 2: Apply the values to Shannon's formula.** $C = B \times \log_2(1 + \text{SNR})$ $C = 1000 \times \log_2(1 + 10,000)$ $C = 1000 \times \log_2(10,001)$
- **Step 3: Calculate the logarithm and solve.** *(Note: $\log_2(10,001) \approx 13.2878$)* $C \approx 1000 \times 13.2878$ $C \approx 13,287.8 \text{ bps}$

Final Answer: The maximum bit rate (capacity) for this channel is approximately **13.29 kbps** (or 13,287.8 bits per second).

You Asked

(g) Evaluate the Ring, Star, and Mesh topology based on the following criteria : 3

(i) Security

(ii) Cost

(iii) Scalability.

Gemini 3.1 Pro

Here is the evaluation of the Ring, Star, and Mesh network topologies based on the requested criteria:

Network Topology Evaluation

Criteria	Ring Topology	Star Topology	Mesh (Full) Topology
(i) Security	**Low:** Data packets travel through intermediate nodes before reaching their destination. Any node on the ring could potentially intercept or read the traffic passing through it.	**Moderate/High:** Security is better because data travels from the sending node to the central device (like a switch) and then directly to the destination. Other nodes do not see the traffic. Security depends primarily on the central hub/switch.	**Highest:** Data travels on dedicated point-to-point links directly between the sender and receiver. There are no intermediate nodes to intercept the data, ensuring privacy and security.
(ii) Cost	**Low:** It requires relatively little cabling since each device only connects to its two immediate neighbors.	**Moderate:** It requires a separate cable from every single device to a central location, plus the cost of purchasing the central hub or switch itself.	**Highest:** Extremely expensive due to the massive amount of cabling required. Every device must have a dedicated physical cable connected to *every other device* in the network, requiring multiple network ports on each computer.

Criteria	Ring Topology	Star Topology	Mesh (Full) Topology
(iii) Scalability	**Poor:** Adding a new device requires physically breaking the ring loop, which temporarily brings down the entire network. Furthermore, adding too many nodes increases the delay for data traveling around the ring.	**High:** Very easy to scale. You can add a new device simply by plugging a single cable from the new computer into the central switch without disrupting the rest of the network (limited only by the number of ports on the central switch).	**Poor:** Highly impractical to scale. Adding just one new device to an existing full mesh network means you have to run a new cable from that device to *every single existing machine* in the entire network.

You Asked

(h) Compare the Virtual Circuit approach and Datagram Network approach based on the following parameters : 3

(i) Quality of Service

(ii) Effect of Router Failure

(iii) Congestion Control.

(i) What type of unguided media is used for the following applications ? 3

(i) Satellite networks

(ii) AM and FM

(iii) Television remotes.

Gemini 3.1 Pro

Here are the answers to the networking questions presented in the image:

(h) Compare Virtual Circuit and Datagram Network approaches

Parameter	Virtual Circuit Approach	Datagram Network Approach
(i) Quality of Service (QoS)	**Easy to guarantee.** Because a dedicated path is established before data transfer begins, resources (like bandwidth and buffers) can be reserved in advance along that path to ensure specific QoS requirements are met.	**Difficult to guarantee.** It uses a "best-effort" delivery model. Packets are routed independently, and no resources are reserved beforehand, meaning delays and jitter can vary unpredictably.
(ii) Effect of Router Failure	**Severe impact.** If a router along the established virtual circuit fails, the entire connection is broken. All subsequent packets will be lost until a brand new circuit is established through a different path.	**Minimal impact.** Because packets are treated independently, if a router fails, the network dynamically updates its routing tables, and subsequent packets are simply routed around the failed node to reach their destination.
(iii) Conges	**Preventative (Admission Control).** Congestion can be avoided during the connection setup phase. If th	**Reactive.** The network cannot prevent packets from entering. When congestion occurs, it must react dyn

Parameter	Virtual Circuit Approach	Datagram Network Approach
tion Control	e network is already congested, it will simply refuse to establish new virtual circuits until traffic clears.	amically, typically by dropping packets or sending choke messages to tell senders to slow down.

(i) Type of unguided media used for applications

Unguided media transport electromagnetic waves without using a physical conductor (wireless communication).

- **(i) Satellite networks: Microwaves** *(Specifically, directional microwaves, which travel in a straight line and require line-of-sight, making them ideal for transmitting up to satellites and back down to earth).*
- **(ii) AM and FM: Radio waves** *(Radio waves are omnidirectional, meaning they propagate in all directions from the antenna, which is why you don't have to point your car antenna directly at the radio station to receive a signal).*
- **(iii) Television remotes: Infrared (IR) waves** *(Infrared is used for short-range communication in closed areas because it relies on line-of-sight and cannot penetrate walls, preventing your remote from changing the channel on your neighbor's TV).*

You Asked

(j) A URL has three components. Extract the three components of the given URL : http://www.abc.india.edu/index.html 3

Gemini 3.1 Pro

Here is the extraction of the three components from the given URL:

Given URL: `http://www.abc.india.edu/index.html`

The three main components are:

1. **Protocol (or Scheme):** `http` *(This indicates the communication rules used to retrieve the document, which in this case is the Hypertext Transfer Protocol).*
2. **Domain Name (or Host):** `www.abc.india.edu` *(This identifies the specific web server on the internet where the resource is hosted).*
3. **Path (or File Name):** `/index.html` *(This specifies the exact location and name of the file being requested on that server).*

You Asked

2. (a) Explain the *three* CSMA protocols. 3

(b) How long does it take for a station to detect a collision in the CSMA/CD protocol ? Justify your answer with the help of a diagram. 4

Gemini 3.1 Pro

Here are the answers to the networking questions presented in the image:

2. (a) Explain the three CSMA protocols.

Carrier Sense Multiple Access (CSMA) is a MAC protocol based on the principle of "listen before you talk." A station senses the medium before transmitting to reduce the chance of collisions. The three main persistence strategies for CSMA are:

1. **1-Persistent CSMA:**
   - **How it works:** When a station has data to send, it continuously senses the channel. If the channel is idle, it transmits immediately (with a probability of 1). If the channel is busy, it keeps listening continuously until it becomes idle, and then immediately transmits.
   - **Pros/Cons:** It has the highest throughput in low-load conditions because stations don't wait unnecessarily. However, in high-load conditions, if multiple stations are waiting for the channel to become idle, they will all transmit simultaneously the moment it frees up, guaranteeing a collision.
2. **Non-Persistent CSMA:**
   - **How it works:** A station senses the channel. If it is idle, it transmits immediately. If it is busy, the station *does not* continuously sense it. Instead, it waits for a random period of time and then senses the channel again.
   - **Pros/Cons:** This significantly reduces the chances of a collision because waiting stations are scattered by their random backoff timers. The drawback is that it introduces longer delays, and the channel might remain idle even if stations have data to send.
3. **p-Persistent CSMA:**
   - **How it works:** This is typically used in slotted channels. When a station has data, it senses the channel. If idle, it transmits with a probability of p. With a probability of $q = (1 - p)$, it defers its transmission to the next time slot. If the next slot is also idle, it either transmits or defers again based on the same probabilities. If the channel becomes busy, it acts as if a collision occurred and waits for a random backoff time.
   - **Pros/Cons:** It attempts to balance the immediate transmission of 1-persistent with the collision reduction of non-persistent CSMA.

---

2. (b) How long does it take for a station to detect a collision in the CSMA/CD protocol?

In CSMA/CD (Carrier Sense Multiple Access with Collision Detection), the worst-case time it takes for a transmitting station to detect a collision is **twice the maximum propagation delay** $(2 \times T_p)$ of the network, also known as the Round-Trip Time (RTT).

$$\text{Maximum Collision Detection Time} = 2 \times T_p$$

Where T_p is the maximum propagation delay (the time it takes for a signal to travel from one end of the network to the farthest opposite end).

Justification:

To understand why it takes $2 \times T_p$, we must look at the worst-case scenario. Imagine a network with two stations, **Station A** and **Station B**, located at the absolute extreme ends of the maximum cable length.

1. **Time $t = 0$:** Station A senses an idle channel and begins transmitting a frame. The signal travels down the cable towards Station B.
2. **Time $t = T_p - \epsilon$:** (Where ϵ is an infinitesimally small amount of time). The signal from A has almost reached B. At this exact moment, Station B decides it wants to transmit. It senses the channel, and because A's signal hasn't *quite* arrived yet, B incorrectly senses the channel as idle and starts its own transmission.
3. **Time $t = T_p$:** A collision occurs immediately at Station B's end of the wire. Station B detects the collision instantly and stops.
4. **The Return Journey:** The resulting collision signal (or jam signal) must now travel all the way back across the entire length of the cable to reach Station A so that A knows its transmission failed.
5. **Time $t = T_p + T_p = 2T_p$:** The collision signal finally arrives at Station A.

Because of this worst-case scenario, a station in a CSMA/CD network must continue transmitting its frame for *at least* $2T_p$. If it finishes transmitting its entire frame before $2T_p$ has elapsed, it might assume the transmission was successful, unaware that a collision signal is currently traveling back towards it. This is why Ethernet has a minimum frame size requirement (e.g., 64 bytes).

You Asked

(c) Define Hamming code. Consider a coding scheme with the following legal codewords : 00010111, 11110010. Calculate its Hamming distance. 2

(d) Compute the bit stream transmitted using Hamming code with even parity for the message M = 1101101. Show the steps to detect and correct an error at the receiver's end if the third bit from the left is inverted during transmission. 6

Gemini 3.1 Pro

Here are the step-by-step solutions to the problems presented in the image:

(c) Hamming Code and Hamming Distance

1. Define Hamming Code: A Hamming code is a type of linear error-correcting code capable of detecting up to two simultaneous bit errors and correcting single-bit errors. It works by strategically inserting redundant "parity bits" into the original message data at specific positions (usually powers of 2). These parity bits ensure that groups of bits within the message adhere to a specific parity rule (either even or odd), allowing the receiver to mathematically pinpoint the exact location of a corrupted bit.

2. Calculate Hamming Distance: The Hamming distance between two legal codewords is the number of bit positions in which they differ. We can find this by performing an XOR (Exclusive OR) operation between the two strings and counting the number of `1` s in the result.

- **Codeword 1:** `00010111`
- **Codeword 2:** `11110010`
- **XOR Result:** `11100101`

Counting the `1` s in the XOR result gives us 5. **Answer:** The Hamming distance for this coding scheme is **5**.

(d) Computing and Correcting a Hamming Code

1. Determine the number of parity bits required: For a message length of $m = 7$ bits (1101101), we use the formula $2^r \geq m + r + 1$ to find the number of parity bits (r).

- If $r = 3$: $2^3 \geq 7 + 3 + 1 \Rightarrow 8 \geq 11$ (False)
- If $r = 4$: $2^4 \geq 7 + 4 + 1 \Rightarrow 16 \geq 12$ (True) We need 4 **parity bits**. The total length of the transmitted code will be $7 + 4 = 11$ bits.

2. Position the Parity and Data Bits: Parity bits (P_1, P_2, P_4, P_8) are placed at positions that are powers of 2 (1, 2, 4, 8). The data bits ($D_3, D_5, D_6, D_7, D_9, D_{10}, D_{11}$) fill the remaining slots from left to right.

Position	1	2	3	4	5	6	7	8	9	10	11
Type	P_1	P_2	D_3	P_4	D_5	D_6	D_7	P_8	D_9	D_{10}	D_{11}
Value	?	?	1	?	1	0	1	?	1	0	1

3. Calculate Even Parity Bits:

- P_1 **checks positions 1, 3, 5, 7, 9, 11:** Data bits are 1, 1, 1, 1, 1 . (Five 1 s). To make the total even, $P_1 = 1$.
- P_2 **checks positions 2, 3, 6, 7, 10, 11:** Data bits are 1, 0, 1, 0, 1 . (Three 1 s). To make the total even, $P_2 = 1$.
- P_4 **checks positions 4, 5, 6, 7:** Data bits are 1, 0, 1 . (Two 1 s). Total is already even, so $P_4 = 0$.
- P_8 **checks positions 8, 9, 10, 11:** Data bits are 1, 0, 1 . (Two 1 s). Total is already even, so $P_8 = 0$.

Transmitted Bit Stream: 1 1 1 0 1 0 1 0 1 0 1

4. Error Detection and Correction: The problem states the third bit from the left is inverted during transmission.

- **Original:** 1 1 1 0 1 0 1 0 1 0 1
- **Received:** 1 1 0 0 1 0 1 0 1 0 1 (The 1 at position 3 became 0)

The receiver recalculates the parity checks (Syndrome bits C_1, C_2, C_4, C_8). If a group has odd parity, the check fails (1).

- C_1 **(1, 3, 5, 7, 9, 11):** 1, 0, 1, 1, 1, 1 . (Five 1 s = Odd) $\Rightarrow \mathbf{C_1 = 1}$
- C_2 **(2, 3, 6, 7, 10, 11):** 1, 0, 0, 1, 0, 1 . (Three 1 s = Odd) $\Rightarrow \mathbf{C_2 = 1}$
- C_4 **(4, 5, 6, 7):** 0, 1, 0, 1 . (Two 1 s = Even) $\Rightarrow \mathbf{C_4 = 0}$
- C_8 **(8, 9, 10, 11):** 0, 1, 0, 1 . (Two 1 s = Even) $\Rightarrow \mathbf{C_8 = 0}$

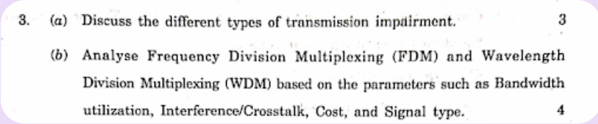
The Syndrome word is $C_8C_4C_2C_1 = 0011$ in binary, which equals **3** in decimal. This correctly identifies that the error occurred at **position 3**. The receiver corrects the stream by flipping the 3rd bit back to 1 .

To help visualize how the parity bits are calculated and how single-bit errors are caught by the syndrome, you can use this interactive tool to explore the Hamming Code logic with your specific data.


```json
{"component":"LlmGeneratedComponent","props":{"height":"600px","prompt":"Objective: Create an interactive Hamming Code Visualizer (Even Parity) to demonstrate encoding and single-bit error correction.\nData State: initialValues = { message: '1101101' }\nStrategy: Form Layout.\nInputs:\n1. A text input to enter a 4 to 11 bit binary message (default to '1101101').\n2. A 'Generate Code' button.\nBehavior:\n- When 'Generate' is clicked, calculate the necessary parity bits and display the 11-bit transmitted string in a grid showing indices 1-11, indicating which are P (parity) and D (data) bits.\n- Below the generated code, display an 'Error Simulation' section. Show the transmitted bits as clickable elements.\n- Allow the user to click exactly one bit to simulate flipping it (injecting an error). Highlight
```

```
the flipped bit.\n- Dynamically display the Syndrome Calculation (C1, C2, C4, C8) based on the current state of the
simulated received bits.\n- Show the final decimal syndrome value and clearly state which bit position the receiver
would identify as the error, proving the correction algorithm works.","id":"im_fbbcb72cb821a17a"}}
```


You Asked


3. (a) Discuss the different types of transmission impairment. 3

(b) Analyse Frequency Division Multiplexing (FDM) and Wavelength Division Multiplexing (WDM) based on the parameters such as Bandwidth utilization, Interference/Crosstalk, Cost, and Signal type. 4

Gemini 3.1 Pro

Here are the detailed answers to the networking questions presented in the image:

3. (a) Types of Transmission Impairment

When a signal travels through a communication medium, it is not perfectly transmitted; the received signal can differ from the original transmitted signal. This degradation is known as transmission impairment. There are three primary types:

1. **Attenuation (Energy Loss):**
    - **What it is:** As a signal travels through a medium, it loses some of its energy in overcoming the resistance of the medium (e.g., electrical signals converting to heat in a wire). This results in a weaker signal at the receiver.
    - **Solution:** Amplifiers are used at intervals along the transmission path to boost the signal back to its original strength.
2. **Distortion (Shape Change):**
    - **What it is:** Distortion alters the actual shape or form of the signal. It occurs most often in composite signals (signals made of multiple frequencies). Because different frequency components travel at slightly different speeds through a medium, they arrive at the receiver at different times, shifting their phases and warping the original signal shape.
    - **Solution:** Equalizers are used to correct phase shifts and restore the signal's shape.
3. **Noise (Unwanted Signals):**
    - **What it is:** Noise is any extraneous, unwanted energy added to the signal during transmission. It is the most significant limiting factor in communication systems.
    - **Common types of noise include:**
        - *Thermal Noise:* Random motion of electrons in a wire creates an extra signal.
        - *Induced Noise:* Interference from motors or appliances acting as transmitting antennas.
        - *Crosstalk:* The magnetic/electrical effect of one wire interfering with an adjacent wire (e.g., hearing another conversation on a telephone line).
        - *Impulse Noise:* Spikes in the signal caused by external forces like lightning or power flaws.

---

3. (b) Analysis of FDM and WDM

Both Frequency Division Multiplexing (FDM) and Wavelength Division Multiplexing (WDM) are analog multiplexing techniques used to combine multiple separate signals into one single transmission medium. However, they operate on different principles and physical media.

Here is an analysis based on the requested parameters:

Parameter	Frequency Division Multiplexing (FDM)	Wavelength Division Multiplexing (WDM)
Basic Principle	Divides the total available bandwidth of a medium into a series of non-overlapping electrical frequency bands, assigning one to each signal.	Combines multiple beams of light, each at a different wavelength (color), onto a single optical fiber. (Conceptually similar to FDM, but applied to light).
Bandwidth Utilization	Moderate to High. However, it requires "guard bands" (empty frequency spaces) between channels to prevent interference, which wastes some of the available bandwidth.	Extremely High. Fiber optic cables have a massive frequency spectrum. WDM (and DWDM - Dense WDM) can carry dozens or hundreds of huge data streams simultaneously, maximizing fiber capacity.
Interference / Crosstalk	Susceptible. If guard bands are too narrow or filters are imprecise, adjacent channels can interfere (crosstalk). Also vulnerable to external Electromagnetic Interference (EMI) if using copper wire.	Very Low. Different wavelengths of light do not easily interfere with each other within the fiber. Completely immune to external Electromagnetic Interference (EMI).
Cost	Generally lower. The required hardware (filters, electronic modulators, demodulators) is mature, widely available, and relatively inexpensive.	High. Requires specialized, highly precise equipment including precise lasers, optical multiplexers (like prisms/diffraction gratings), and specialized optical amplifiers.
Signal Type	Used primarily for **Analog** electrical or radio signals (though digital data can be carried if modulated into an analog signal first).	Used exclusively for **Optical** signals (light waves) traveling through fiber-optic cables.

You Asked

(c) The following character encoding is used in a data link layer protocol : 8

A : 11010101

B : 10101001

FLAG : 01111110

ESC : 10100011

Show the bit sequence transmitted (in binary) for the five-character frame A ESC B ESC FLAG when each of the following framing methods is used :

(*i*) Character count

(*ii*) Flag bytes with byte stuffing

(*iii*) Starting and ending flag bytes with bit stuffing.

Also, find the original data for the given output obtained after applying the byte-stuffing technique :

FLAG A B ESC ESC C ESC FLAG ESC FLAG D E FLAG

Gemini 3.1 Pro

Here are the step-by-step solutions to the data link layer framing questions based on the provided character encodings:

Given Encodings:

- **A:** `11010101`
- **B:** `10101001`
- **FLAG:** `01111110`
- **ESC:** `10100011`

Original Payload: `A ESC B ESC FLAG`

(i) Character Count

Rule: The frame begins with a single byte that indicates the total number of characters in the frame, including the count byte itself. No stuffing is performed.

1. **Count:** The payload has 5 characters (`A`, `ESC`, `B`, `ESC`, `FLAG`). Adding the count byte itself brings the total to **6**.
2. **Binary for 6:** `00000110`
3. **Frame Structure:** `[COUNT=6] A ESC B ESC FLAG`

Transmitted Bit Sequence: `00000110 11010101 10100011 10101001 10100011 01111110`

(ii) Flag bytes with byte stuffing

Rule: The frame is encapsulated between starting and ending `FLAG` bytes. To prevent payload data from accidentally being read as a control character, an `ESC` byte is inserted (stuffed) immediately before any `ESC` or `FLAG` byte that appears natively in the payload.

1. **Original Payload:** `A`, `ESC`, `B`, `ESC`, `FLAG`
2. **Apply Stuffing:**
   - `A` $\rightarrow$ remains `A`
   - `ESC` $\rightarrow$ becomes `ESC ESC`
   - `B` $\rightarrow$ remains `B`
   - `ESC` $\rightarrow$ becomes `ESC ESC`
   - `FLAG` $\rightarrow$ becomes `ESC FLAG`
3. **Stuffed Payload:** `A ESC ESC B ESC ESC ESC FLAG`
4. **Add Frame Boundaries:** `FLAG [Stuffed Payload] FLAG`

Transmitted Bit Sequence:
`01111110 11010101 10100011 10100011 10101001 10100011 10100011 10100011 01111110 01111110`

(iii) Starting and ending flag bytes with bit stuffing

Rule: The frame starts and ends with the `01111110` flag. The payload is treated as a continuous stream of bits. To ensure the flag pattern never accidentally appears in the data, a `0` is stuffed immediately after any five consecutive `1`s in the payload stream.

1. **Payload Binary:** `11010101 10100011 10101001 10100011 01111110`

2. **Scan for five consecutive 1s:** The only occurrence of five consecutive `1`s is within the final `FLAG` character of the payload: `...01111110`
3. **Apply Bit Stuffing:** Insert a `0` after the fifth `1`. `...011111` **`0`** `10`
4. **Stuffed Payload Stream:** `11010101 10100011 10101001 10100011 011111010`
5. **Add Frame Boundaries:** Add `01111110` to the start and end.

Transmitted Bit Sequence: `01111110 11010101 10100011 10101001 10100011 011111010 01111110`

Reverse Byte-Stuffing (De-stuffing)

Given Received Frame: `FLAG A B ESC ESC C ESC FLAG ESC FLAG D E FLAG`

Rule for Receiver:

- The first and last `FLAG`s represent the frame boundaries and are removed.
- When an `ESC` byte is encountered in the data, the receiver discards it and accepts the *immediately following byte* as regular data, regardless of what it is.

De-stuffing Process:

1. `FLAG` → (Discard: Start Frame)
2. `A` → Data: **A**
3. `B` → Data: **B**
4. `ESC ESC` → (Discard first ESC) → Data: **ESC**
5. `C` → Data: **C**
6. `ESC FLAG` → (Discard first ESC) → Data: **FLAG**
7. `ESC FLAG` → (Discard first ESC) → Data: **FLAG**
8. `D` → Data: **D**
9. `E` → Data: **E**
10. `FLAG` → (Discard: End Frame)

Original Data Payload: `A B ESC C FLAG FLAG D E`

You Asked

4. (a) "Distance Vector Routing algorithm reacts rapidly to good news but leisurely to bad news". Justify the statement. 2

Gemini 3.1 Pro

4. (a) Distance Vector Routing: "Good News Travels Fast, Bad News Travels Slow"

This statement refers to a fundamental flaw in basic Distance Vector Routing (DVR) algorithms, known as the **Count-to-Infinity problem**. Here is the justification:

1. Good News Travels Fast (Rapid Reaction to Good News): When a new, highly efficient link is established or a router comes online (good news), this information propagates quickly through the network.

- **Mechanism:** If Router A discovers a new path to a destination with a lower cost (e.g., cost goes from 10 down to 2), it immediately updates its own routing table. During its next scheduled update cycle, it broadcasts this new, lower cost to its neighbors. Because the new cost is better than what the neighbors currently have, they immediately accept the update and adjust their own tables. This cascade happens efficiently and quickly across the network.

2. Bad News Travels Slowly (Leisurely Reaction to Bad News): When a link fails entirely or a cost increases dramatically (bad news), DVR struggles to disseminate this information efficiently, leading to routing loops.

- **Mechanism:** Imagine a simple path: `A --- B --- C`.
  - Router A reaches C via B (Cost: 2).
  - Router B reaches C directly (Cost: 1).
  - Now, the link between B and C fails (Cost: ∞).
  - B detects the failure and updates its cost to C as ∞.
  - **The Problem:** Before B can broadcast its new ∞ cost, Router A sends its regular update to B saying, "Hey, I can reach C with a cost of 2!"
  - Router B, not knowing that A's path to C actually goes *through B itself*, mistakenly believes A has an alternative route. B updates its table: "I can reach C via A with a cost of $2 + 1 = 3$."
  - During the next cycle, B tells A its new cost is 3. A updates its cost to $3 + 1 = 4$.
  - They continue bouncing these updates back and forth (A cost: 6, B cost: 7... A cost: 10, B cost: 11...), slowly counting upwards toward infinity until the metric reaches the protocol's predefined maximum limit (e.g., 16 in RIP), finally recognizing the route is dead.

This slow convergence process is why bad news travels leisurely in distance vector networks.

You Asked

(*b*) Consider a network of six routers labelled A, B, C, D, E and F. The cost of the links between the routers is as follows : 5

(*i*) A-B : 4

(*ii*) A-E : 5

(*iii*) B-C : 2

(*iv*) B-F : 6

(*v*) C-D : 3

(*vi*) C-E : 1

(*vii*) D-F : 7

(*viii*) F-E : 8

The following information has just arrived at the router C :

- From B : (5, 0, 8, 12, 6, 2)
- From D : (16, 12, 6, 0, 9, 10)
- From E : (7, 6, 3, 9, 0, 4)

The measured delays to B, D and E are 6, 3 and 5 respectively. Using distance vector routing algorithm, give the new routing table for C specifying both the delay and the outgoing line to use.

Gemini 3.1 Pro

Based on the information provided in the image, here is the step-by-step calculation to find the new routing table for router C using the Distance Vector Routing (Bellman-Ford) algorithm.

1. Identify the Given Data

- **Router C's Neighbors:** B, D, and E.
- **Measured delays to neighbors:**
  - Cost to B: $c(C, B) = 6$
  - Cost to D: $c(C, D) = 3$
  - Cost to E: $c(C, E) = 5$
- **Received Distance Vectors (Assuming destinations are A, B, C, D, E, F):**
  - From B (D_B): `[5, 0, 8, 12, 6, 2]`
  - From D (D_D): `[16, 12, 6, 0, 9, 10]`
  - From E (D_E): `[7, 6, 3, 9, 0, 4]`

2. Apply the Routing Algorithm

The routing algorithm dictates that router C will calculate its new minimum delay to a destination x by checking the path through each of its neighbors using the formula: $D_C(x) = \min\{c(C, v) + D_v(x)\}$ *(where v is the neighbor B, D, or E)*

Calculations for each destination:

- **To Destination A:**
  - Via B: $6 + 5 = 11$
  - Via D: $3 + 16 = 19$
  - Via E: $5 + 7 = 12$
  - *Minimum is **11** (Outgoing line: **B**)*
- **To Destination B:**
  - Via B: $6 + 0 = 6$
  - Via D: $3 + 12 = 15$
  - Via E: $5 + 6 = 11$
  - *Minimum is **6** (Outgoing line: **B**)*
- **To Destination C (Self):**
  - Distance to itself is always **0**. (Outgoing line: **Local / -**)
- **To Destination D:**
  - Via B: $6 + 12 = 18$
  - Via D: $3 + 0 = 3$
  - Via E: $5 + 9 = 14$
  - *Minimum is **3** (Outgoing line: **D**)*
- **To Destination E:**

- Via B: $6 + 6 = 12$
- Via D: $3 + 9 = 12$
- Via E: $5 + 0 = 5$
- *Minimum is **5** (Outgoing line: **E**)*

- **To Destination F:**
  - Via B: $6 + 2 = 8$
  - Via D: $3 + 10 = 13$
  - Via E: $5 + 4 = 9$
  - *Minimum is **8** (Outgoing line: **B**)*

---

3. Final Routing Table for C

Here is the updated routing table specifying both the delay and the outgoing line:

Destination	New Delay	Outgoing Line
A	11	B
B	6	B
C	0	-
D	3	D
E	5	E
F	8	B

You Asked

(c) How IPv6 is more secure than the IPv4 protocol ? Discuss the various fields of IPv6 headers with the help of a diagram. Why is the Header checksum of an IPv4 packet computed at every hop from source to destination ? 8

Gemini 3.1 Pro Extended

Here are the detailed answers to the networking questions presented in the image:

1. How IPv6 is more secure than the IPv4 protocol

IPv6 was designed with security in mind from the ground up, whereas security in IPv4 was largely an afterthought. The primary reasons IPv6 is considered more secure include:

- **Native IPsec Support:** IPsec (Internet Protocol Security) provides end-to-end encryption and authentication. In IPv4, IPsec is an optional add-on that requires careful configuration and often struggles with NAT (Network Address Translation). In

IPv6, IPsec support is a mandatory part of the protocol suite (implemented via Extension Headers). It inherently supports data confidentiality (ESP - Encapsulating Security Payload) and data integrity/authentication (AH - Authentication Header).

- **Vast Address Space:** IPv6 uses 128-bit addresses compared to IPv4's 32-bit addresses. This makes malicious network scanning and reconnaissance (like ping sweeps or port scans to find vulnerable hosts) practically impossible. Scanning a standard IPv6 `/64` subnet would take millions of years at current speeds.
- **Secure Neighbor Discovery (SEND):** IPv4 relies on ARP (Address Resolution Protocol), which is highly vulnerable to ARP spoofing and poisoning attacks. IPv6 replaces ARP with Neighbor Discovery Protocol (NDP). It includes an extension called SEND, which uses cryptographic techniques to prevent malicious hosts from intercepting or redirecting traffic on the local network.

---

2. Discuss the various fields of IPv6 headers with a diagram

The IPv6 base header is streamlined to be exactly 40 bytes long, allowing routers to process packets faster than the variable-length IPv4 header.

Here is a breakdown of the fields, which you can explore in the interactive diagram below:

- **Version (4 bits):** Always set to `6` to indicate IPv6.
- **Traffic Class (8 bits):** Similar to the Type of Service (ToS) field in IPv4. It is used for Quality of Service (QoS) to prioritize specific types of traffic (e.g., voice or video).
- **Flow Label (20 bits):** Used to maintain the sequential flow of packets belonging to a specific communication. It tells routers to keep packets of the same flow on the same path to prevent reordering delays.
- **Payload Length (16 bits):** Indicates the size of the payload in bytes (this includes any Extension Headers plus the upper-layer data like TCP/UDP).
- **Next Header (8 bits):** Replaces the IPv4 "Protocol" field. It indicates what type of header immediately follows the basic IPv6 header (e.g., TCP, UDP, or an IPv6 Extension Header like IPsec).
- **Hop Limit (8 bits):** Replaces the IPv4 "Time to Live" (TTL) field. It is decremented by 1 at each router. If it reaches zero, the packet is discarded.
- **Source Address (128 bits):** The IP address of the sender.
- **Destination Address (128 bits):** The IP address of the intended recipient.


```json
{"component":"LlmGeneratedComponent","props":{"height":"650px","prompt":"Objective: Create an interactive diagram of the IPv6 Base Header to explain its various fields.\nData State: initialValues = { selectedField: null }\nStrategy: Form Layout. Interactive Educational Diagram.\nInputs:\n1. A visual grid representing the 32-bit wide IPv6 header structure. The rows should represent 32-bit (4-byte) increments.\n- Row 1: Version (4 bits), Traffic Class (8 bits), Flow Label (20 bits)\n- Row 2: Payload Length (16 bits), Next Header (8 bits), Hop Limit (8 bits)\n- Rows 3-6: Source Address (128 bits, spans 4 rows)\n- Rows 7-10: Destination Address (128 bits, spans 4 rows)\nThese grid sections must be clickable buttons/regions.\nBehavior:\n- Render the header as a clear, proportional grid (e.g., Version takes 1/8th of the width, Flow Label takes 5/8ths, Source Address spans full width for 4 rows).\n- When the user clicks a specific field in the diagram, highlight that field visually to distinguish it.\n- Below the diagram, display an information panel that dynamically updates based on the selected field. Show the Field Name, Size in Bits, and a brief Description based on standard IPv6 networking definitions.\n- Include a prompt at the top saying 'Click on any field in the IPv6 header to learn more about it.'","id":"im_3af13487bdc077f9"}}
```


---

3. Why is the Header Checksum of an IPv4 packet computed at every hop?

The IPv4 header includes a field called **Time to Live (TTL)**. The purpose of the TTL is to prevent packets from looping endlessly in the network if there is a routing error.

Every single time a router receives an IPv4 packet, it **must decrement the TTL value by 1**. Because the TTL field is physically located inside the IPv4 header, the actual binary data of the header changes at every router.

Since the Header Checksum is a mathematical calculation based on the bits within the header, altering the TTL invalidates the original checksum. Therefore, every router along the path from source to destination must recalculate and update the Header Checksum before forwarding the packet to ensure the next router doesn't discard it as corrupted.

(Note: To speed up routing, IPv6 completely eliminated the Header Checksum field. It relies on upper-layer protocols like TCP and lower-layer protocols like Ethernet to handle error checking, making IPv6 routers more efficient).

You Asked

5. (a) Explain the working of Stop and Wait protocol. 4

(b) What kind of duplexing mode is used in the following applications : 3

(i) Voice over IP

(ii) Walkie-Talkie

(iii) Telephone.

Gemini 3.1 Pro Extended

Here are the answers to the networking questions presented in the image:

5. (a) Working of the Stop and Wait Protocol

The Stop and Wait protocol is the simplest form of flow control and error control (when implemented as Stop-and-Wait ARQ) used in the Data Link Layer. It ensures that a sender does not overwhelm a receiver with data by forcing the sender to wait for confirmation before sending more.

Here is the step-by-step working:

1. **Transmission:** The sender transmits a single frame of data to the receiver and immediately starts a countdown timer.
2. **Stop and Wait:** After sending the frame, the sender completely stops transmitting and waits for an acknowledgment (ACK) from the receiver.
3. **Reception & Acknowledgment:** If the receiver successfully receives the frame and it is error-free, it processes the data and sends an ACK frame back to the sender.
4. **Next Frame:** If the sender receives the ACK before its timer expires, it resets the timer and transmits the next frame in the sequence.
5. **Error Handling (Timeouts):**
    - **Lost Frame:** If the transmitted frame is lost in transit, the receiver never gets it and sends no ACK. The sender's timer will eventually expire (timeout). The sender assumes the frame was lost and **retransmits** the original frame.
    - **Lost ACK:** If the frame arrives, but the returning ACK is lost, the sender's timer will still expire. The sender retransmits the frame. The receiver will recognize this as a duplicate (using sequence numbers 0 and 1) and discard the duplicate payload, but it will still send another ACK to satisfy the sender.

To help you visualize how the sender and receiver interact during these different scenarios (including timeouts and data loss), you can use this interactive simulator.

json


```json
{"component":"LlmGeneratedComponent","props":{"height":"600px","prompt":"Objective: Create an interactive Stop-and-Wait ARQ protocol visualizer to demonstrate normal operation, lost frames, and lost ACKs.\nData State: initialValues = {}\nStrategy: Standard Layout. Use Anime.js to animate the flow of data packets between a Sender node and a Receiver node.\nInputs:\n1. Three action buttons: 'Normal Transmission', 'Simulate Lost Frame', and 'Simulate Lost ACK'.\nBehavior:\n- Display a 'Sender' node on the left and a 'Receiver' node on the right, separated by a wide space representing the network.\n- Include a visual 'Timer' element next to the Sender.\n- Normal Transmission: Animate a data packet moving from Sender to Receiver. Once it arrives, animate an ACK packet moving back to the Sender. Display status text: 'Frame received. ACK sent.'\n- Lost Frame: Animate a data packet moving from Sender, but fade it out halfway across the network. Start the Sender's timer. When the timer runs out, display 'Timeout!' and automatically animate a retransmission of the packet that successfully completes the round trip.\n- Lost ACK: Animate a data packet reaching the Receiver. The Receiver generates an ACK, but the ACK fades out halfway back to the Sender. The Sender's timer runs out, displays 'Timeout!', and retransmits the frame. The Receiver displays 'Duplicate frame discarded, resending ACK' and successfully sends the ACK back.\n- Keep animations smooth and educational, clearly labeling packets as 'Data 0', 'ACK 1', etc., to show sequence numbers if possible.","id":"im_caf7660147374885"}}
```

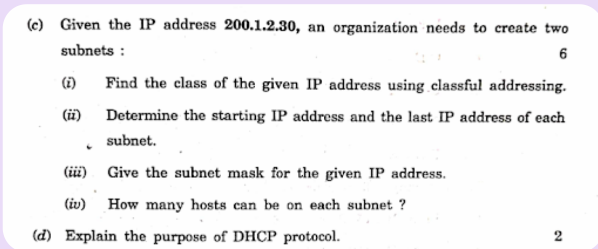

---

5. (b) Duplexing Modes in Applications

Duplexing modes describe the direction and timing of data transmission between two devices.

- **(i) Voice over IP (VoIP): Full-Duplex**
  - *Reasoning:* VoIP allows both parties on the call to speak and hear each other simultaneously, exactly like a standard telephone call. Data travels in both directions at the same time.
- **(ii) Walkie-Talkie: Half-Duplex**
  - *Reasoning:* A walkie-talkie allows two-way communication, but not at the exact same time. A user must press a "push-to-talk" button to transmit, during which time they cannot hear the other person. Only one direction is active at any given moment.
- **(iii) Telephone: Full-Duplex**
  - *Reasoning:* Standard telecommunication networks (whether landline or cellular) are designed to let both the caller and receiver talk and listen simultaneously without cutting each other off.

You Asked



(c) Given the IP address **200.1.2.30**, an organization needs to create two subnets : 6

(i) Find the class of the given IP address using classful addressing.

(ii) Determine the starting IP address and the last IP address of each subnet.

(iii) Give the subnet mask for the given IP address.

(iv) How many hosts can be on each subnet ?

(d) Explain the purpose of DHCP protocol. 2

Gemini 3.1 Pro Extended

Here are the step-by-step solutions to the networking questions presented in the image:

(c) Subnetting the IP Address 200.1.2.30

The given IP address is `200.1.2.30`. The organization wants to create **two subnets**.

(i) Find the class of the given IP address using classful addressing. To determine the class, we look at the first octet of the IP address, which is **200**.

- Class A: 1 – 126
- Class B: 128 – 191
- Class C: 192 – 223 Because 200 falls in the 192-223 range, this is a **Class C** IP address. The default network identifier is `200.1.2.0` with a default subnet mask of 255.255.255.0.

(iii) Give the subnet mask for the given IP address. *(Solving this first makes step (ii) easier)* To divide the default Class C network into 2 subnets, we must borrow bits from the host portion.

- Formula for subnets: $2^n = \text{Number of subnets}$ (where n is bits borrowed).
- $2^1 = 2$. Therefore, we need to borrow **1 bit**.
- The default Class C mask is `/24` (24 network bits). Borrowing 1 bit makes the new subnet mask **`/25`** (25 network bits).
- In binary: `11111111.11111111.11111111.10000000`
- In decimal, the new custom subnet mask is **255.255.255.128**.

(iv) How many hosts can be on each subnet? After borrowing 1 bit for the subnet, we have 7 bits left for the hosts (since an IPv4 address is 32 bits, $32 - 25 = 7$).

- Formula for usable hosts: $2^h - 2$ (where h is the remaining host bits).
- $2^7 - 2 = 128 - 2 = \mathbf{126}$
- Each subnet can have **126** usable hosts. (We subtract 2 because the first address is reserved for the Network ID and the last for the Broadcast ID).

(ii) Determine the starting IP address and the last IP address of each subnet. With a block size of 128 (determined by the borrowed bit), the two subnets are mapped out as follows:

- **Subnet 1:**
  - Network ID: `200.1.2.0`
  - **Starting Usable IP: 200.1.2.1**
  - **Last Usable IP: 200.1.2.126**
  - Broadcast ID: `200.1.2.127`
- **Subnet 2:**
  - Network ID: `200.1.2.128`
  - **Starting Usable IP: 200.1.2.129**
  - **Last Usable IP: 200.1.2.254**
  - Broadcast ID: `200.1.2.255`

To help you visualize how borrowing host bits changes the block sizes, usable host ranges, and subnet masks, you can use this interactive subnetting tool.

```
json
```

{"component":"LlmGeneratedComponent","props":{"height":"600px","prompt":"Objective: Create an interactive IPv4 Subnet Calculator to visualize splitting a network into smaller subnets.\nData State: initialValues = { baseIp: '200.1.2.0', defaultCidr: 24, targetCidr: 25 }\nStrategy: Form Layout.\nInputs:\n1. Text input for Base Network IP (default '200.1.2.0').\n2. Dropdown or Number Input for Starting CIDR prefix (default /24).\n3. Dropdown or Number Input for Target CIDR prefix (default /25, must be >= Starting CIDR).\nBehavior:\n- Calculate the number of created subnets (2^(Target - Starting)).\n- Calculate the new Subnet Mask in dotted decimal format based on the Target CIDR.\n- Calculate the number of usable hosts per subnet (2^(32 - Target) - 2).\n- Display a table listing each generated subnet (limit rendering to the first 16 subnets to avoid performance issues if a very large subnet count is selected). For each subnet, show the Subnet Number, Network Address, First Usable IP, Last Usable IP, and Broadcast Address.\n- Update the table and calculation readouts dynamically when any input changes.","id":"im_bbd3f225cdaebb4d"}}

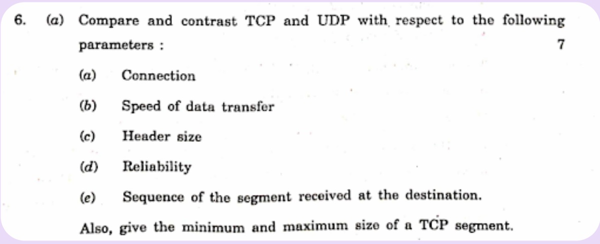
(d) Purpose of the DHCP Protocol

DHCP (Dynamic Host Configuration Protocol) is a network management protocol used to automate the process of configuring devices on an IP network.

Its primary purpose is to **automatically assign IP addresses** and other essential network parameters (such as the subnet mask, default gateway, and DNS server addresses) to client devices (like computers, smartphones, or printers) when they connect to the network.

This automation eliminates the need for network administrators to manually configure static IP addresses on every single device, significantly reducing administrative overhead and preventing configuration errors like IP address conflicts (where two devices accidentally receive the same IP address).

You Asked


6. (a) Compare and contrast TCP and UDP with respect to the following parameters : 7

(a) Connection

(b) Speed of data transfer

(c) Header size

(d) Reliability

(e) Sequence of the segment received at the destination.

Also, give the minimum and maximum size of a TCP segment.

Gemini 3.1 Pro Extended

Here are the answers to the networking question presented in the image:

6. (a) Compare and Contrast TCP and UDP

Here is the comparison between the Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) based on the specified parameters:

Parameter	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
(a) Connection	**Connection-oriented:** A logical connection (3-way handshake) must be established before any data c an be transferred.	**Connectionless:** Data is sent immediately wit hout establishing a prior connection or path.

Parameter	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
(b) Speed of data transfer	**Slower:** The overhead of establishing connections, acknowledging received packets, and performing error/flow control makes it relatively slow.	**Faster:** The absence of connection setup and acknowledgments allows for very rapid data transmission.
(c) Header size	**20 to 60 bytes:** The header is variable and much larger due to fields needed for sequencing, acknowledgments, and control flags.	**8 bytes:** The header is fixed and very small, containing only source/destination ports, length, and a checksum.
(d) Reliability	**Highly Reliable:** Guarantees delivery. It tracks lost or corrupted packets and retransmits them until successfully received.	**Unreliable ("Best-Effort"):** Does not guarantee delivery. If a packet is lost in transit, it is simply gone; there is no retransmission.
(e) Sequence of the segment received	**Ordered:** Uses sequence numbers to guarantee that data is reassembled and delivered to the application in the exact order it was sent.	**Unordered:** Packets travel independently and may arrive at the destination out of sequence. The protocol does not reorder them.

Minimum and Maximum Size of a TCP Segment

- **Minimum Size: 20 bytes**
  - *Explanation:* The minimum size of a TCP segment occurs when it carries no data payload (e.g., a simple acknowledgment or control packet). In this case, the segment consists entirely of the mandatory base TCP header, which is 20 bytes long.
- **Maximum Size: 65,515 bytes**
  - *Explanation:* The absolute maximum size of an IP datagram is 65,535 bytes (defined by the 16-bit total length field in the IPv4 header). To find the maximum TCP segment size, you must subtract the minimum IPv4 header size (20 bytes) from this total. Therefore, $65,535 - 20 = 65,515$ bytes. *(Note: This 65,515-byte segment includes the TCP header. The maximum TCP data payload or MSS is $65,515 - 20 = 65,495$ bytes).*

You Asked

(b) Discuss the various request methods provided by HTTP. 4

(c) Ravi writes an email to his friend and clicks the "Send" button. During this process, a protocol is responsible for transferring Ravi's message from his mail server to the recipient's mail server over the internet. Which protocol is involved in this transfer, and how does it facilitate communication between the mail servers ? 4

Gemini 3.1 Pro Extended

Here are the answers to the networking questions presented in the image:

(b) Various Request Methods Provided by HTTP

HTTP (Hypertext Transfer Protocol) defines a set of request methods (often referred to as HTTP verbs) to indicate the desired action to be performed on a given resource. Here are the primary methods:

- **GET:** The most common method. It is used to request data from a specified resource. GET requests should only retrieve data and have no other effect (they are safe and idempotent).
- **POST:** Used to send data to a server to create or update a resource. The data sent to the server is stored in the request body of the HTTP request (e.g., submitting a web form).
- **PUT:** Used to send data to a server to create or completely replace a target resource with the uploaded content.
- **PATCH:** Similar to PUT, but used to apply *partial* modifications to a resource rather than replacing it entirely.
- **DELETE:** Used to delete the specified resource from the server.
- **HEAD:** Functions exactly like a GET request, but it asks the server to return only the headers and not the actual document body. It's useful for checking what a GET request will return (like checking file size or last modified date) without downloading the whole file.
- **OPTIONS:** Used to describe the communication options (like which HTTP methods are supported) for the target resource.

(c) Email Transfer Protocol (Ravi's Email)

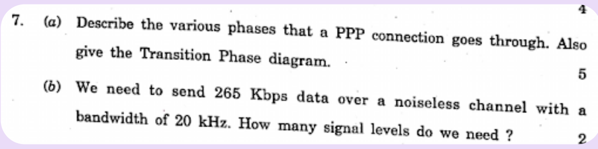
1. Which protocol is involved? The protocol responsible for transferring Ravi's message from his mail server to the recipient's mail server over the internet is **SMTP (Simple Mail Transfer Protocol)**.

2. How does it facilitate communication between the mail servers? SMTP is a "push" protocol used specifically for sending emails. It facilitates communication through the following process:

- **Connection Establishment:** Ravi's mail server (acting as the SMTP client) initiates a reliable TCP connection (typically on port 25) with the recipient's mail server (acting as the SMTP server).
- **Handshake & Commands:** Once connected, the servers communicate using simple, text-based commands.
  - It starts with a greeting (`HELO` or `EHLO`).
  - The sending server specifies who the email is from (`MAIL FROM`).
  - It specifies who the email is going to (`RCPT TO`). If the receiving server accepts the recipient, it acknowledges it.
- **Data Transfer:** The sending server issues the `DATA` command and then transmits the actual email content (headers and body). It signals the end of the message with a single period (`.`) on a new line.
- **Connection Termination:** After successful transmission, the sending server issues the `QUIT` command, and the TCP connection is cleanly closed.

(Note: Protocols like POP3 or IMAP are used by the recipient to "pull" or retrieve the email from their own server to their device, but SMTP handles the actual server-to-server transit).

You Asked


7. (a) Describe the various phases that a PPP connection goes through. Also give the Transition Phase diagram. 4

(b) We need to send 265 Kbps data over a noiseless channel with a bandwidth of 20 kHz. How many signal levels do we need ? 2

Gemini 3.1 Pro Extended

Here are the answers to the networking questions presented in the image:

7. (a) PPP Connection Phases

The Point-to-Point Protocol (PPP) goes through a specific sequence of states to establish, maintain, and terminate a connection.

The phases are:

1. **Dead (Inactive):** This is the default state. The link is not active, and there is no physical layer carrier present. When a physical connection is detected (e.g., a modem connects), it transitions to the Establish phase.
2. **Establish:** In this phase, the Link Control Protocol (LCP) negotiates basic connection options, such as the maximum frame size, authentication protocols, and link quality monitoring. If negotiation is successful, it moves to the next phase.
3. **Authenticate (Optional):** If an authentication protocol (like PAP or CHAP) was agreed upon during the Establish phase, the two parties verify each other's identities here. If authentication fails, the connection drops to the Terminate phase. If it succeeds (or wasn't required), it moves to the Network phase.
4. **Network:** The Network Control Protocols (NCPs) take over to configure the specific network layer protocols being used over the link (usually IP). For example, IPCP (Internet Protocol Control Protocol) is used to assign dynamic IP addresses.
5. **Open:** Once network configuration is complete, the connection is fully open. This is the data transfer phase where actual payload data (user traffic) is exchanged. It remains in this state until one side initiates termination or the physical link drops.
6. **Terminate:** When the connection needs to be closed (either by user request, authentication failure, or a link fault), LCP packets are exchanged to gracefully tear down the logical link. Once finished, the connection returns to the Dead state.

---

7. (b) Signal Levels for a Noiseless Channel

To find the maximum bit rate of a **noiseless** channel, we use the **Nyquist Bit Rate formula**:

$$BitRate = 2 \times Bandwidth \times \log_2(L)$$

Where:

- **BitRate** = Maximum data rate in bits per second (bps)
- **Bandwidth** = Channel bandwidth in Hertz (Hz)
- **L** = Number of signal levels used to represent data

1. Identify the given values:

- $BitRate = 265 \text{ Kbps} = 265,000 \text{ bps}$
- $Bandwidth = 20 \text{ kHz} = 20,000 \text{ Hz}$

2. Set up the equation: $265,000 = 2 \times 20,000 \times \log_2(L)$ $265,000 = 40,000 \times \log_2(L)$

3. Solve for L: Divide both sides by 40,000: $\log_2(L) = \frac{265,000}{40,000}$ $\log_2(L) = 6.625$

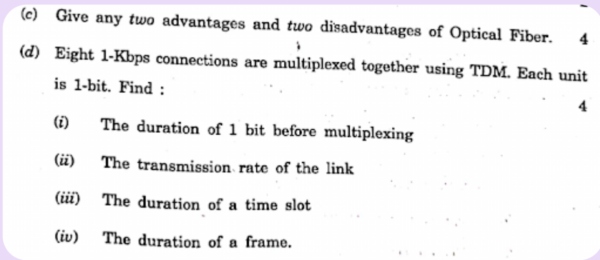
To isolate L, calculate 2 to the power of 6.625: $L = 2^{6.625}$ $L \approx 98.66$

Final Answer: Since the number of signal levels must be an integer, you would strictly need at least **99 signal levels** to achieve a 265 Kbps data rate over a 20 kHz noiseless channel. However, in practical digital communication systems, signal levels are almost always implemented as powers of 2. Therefore, a real-world system would need to jump up to the next power of 2, which is 2^7 or **128 signal levels**.

You can use this interactive calculator to see how changing the bandwidth or the number of signal levels affects the maximum possible bit rate in a noiseless environment according to Nyquist's theorem.


```json
{"component":"LlmGeneratedComponent","props":{"height":"600px","prompt":"Objective: Create an interactive Nyquist Bit Rate Calculator to visualize the relationship between Bandwidth, Signal Levels, and maximum Bit Rate in a noiseless channel.\nData State: initialValues = { bandwidth: 20000, targetBitRate: 265000 }\nStrategy: Form Layout.\nInputs:\n1. Number input for 'Bandwidth (Hz)' (default 20000).\n2. Number input for 'Number of Signal Levels (L)' (default 128).\nBehavior:\n- Display the Nyquist formula: Bit Rate = 2 * Bandwidth * log2(L).\n- Dynamically calculate and display the resulting 'Maximum Bit Rate (bps)' as the user changes the Bandwidth or Signal Levels.\n- Also include a 'Reverse Calculator' section below: Allow the user to input a Target Bit Rate (default 265000) and Bandwidth (default 20000), and dynamically calculate the exact Minimum Signal Levels required, as well as the 'Practical Power of 2 Signal Levels' (rounding up to the nearest power of 2).","id":"im_0089adcfba6988f5"}}
```


You Asked


(c) Give any *two* advantages and *two* disadvantages of Optical Fiber. 4

(d) Eight 1-Kbps connections are multiplexed together using TDM. Each unit is 1-bit. Find : 4

(i) The duration of 1 bit before multiplexing

(ii) The transmission rate of the link

(iii) The duration of a time slot

(iv) The duration of a frame.

Gemini 3.1 Pro Extended

Here are the answers to the networking questions presented in the image:

(c) Advantages and Disadvantages of Optical Fiber

Advantages:

1. **Immense Bandwidth:** Optical fibers can carry significantly more data than copper cables. They support extremely high data transmission rates over long distances, making them ideal for backbone networks.
2. **Immunity to Electromagnetic Interference (EMI):** Because they use light instead of electrical currents to transmit data, optical fibers are completely immune to interference from power lines, radio signals, or lightning. They also do not emit electromagnetic noise.

Disadvantages:

1. **Fragility and Installation Complexity:** The glass core of an optical fiber is fragile. It can be easily broken if bent too sharply (exceeding its bend radius). Furthermore, splicing and terminating fiber optic cables requires highly specialized, expensive equipment and skilled technicians.

2. **High Cost:** While the cost of the cable itself has dropped, the overall system cost (including specialized optical transmitters, receivers, and the labor required for careful installation) remains higher than traditional copper infrastructure.

(d) Time Division Multiplexing (TDM) Calculation

Given Data:

- Number of connections (N) = 8
- Data rate of each connection = $1 \text{ Kbps} = 1000 \text{ bps}$
- Multiplexing unit size = 1 bit

Calculations:

(i) The duration of 1 bit before multiplexing The duration of a single bit on an input line is the inverse of that line's data rate. $\text{Bit Duration} = \frac{1}{\text{Data Rate}}$ $\text{Bit Duration} = \frac{1}{1000 \text{ bps}} = 0.001 \text{ seconds}$ **Answer:** The duration of 1 bit before multiplexing is 1 ms (millisecond).

(ii) The transmission rate of the link To keep up with the incoming data without causing bottlenecks, the high-speed multiplexed link must transmit data at a rate equal to the sum of all input rates. $\text{Link Rate} = N \times \text{Input Data Rate}$ $\text{Link Rate} = 8 \times 1 \text{ Kbps}$ **Answer:** The transmission rate of the link is 8 Kbps (or 8000 bps).

(iii) The duration of a time slot Since the multiplexing unit is 1 bit, one time slot on the main link holds exactly 1 bit. The duration of this slot is the inverse of the high-speed link's transmission rate. $\text{Slot Duration} = \frac{1}{\text{Link Rate}}$ $\text{Slot Duration} = \frac{1}{8000 \text{ bps}} = 0.000125 \text{ seconds}$ **Answer:** The duration of a time slot is $125 \text{ \mus}$ (microseconds). *(Note: This is exactly 1/8th of the original bit duration, as the link is running 8 times faster).*

(iv) The duration of a frame A single TDM frame carries exactly one unit (in this case, 1 bit) from every single connection. Therefore, a frame contains 8 time slots. $\text{Frame Duration} = N \times \text{Slot Duration}$ $\text{Frame Duration} = 8 \times 125 \text{ \mus} = 1000 \text{ \mus}$ **Answer:** The duration of a frame is 1 ms. *(Note: In basic TDM, the duration of a frame on the output link is always exactly equal to the duration of one multiplexing unit on an input link).*

To help you visualize how data from multiple slow lines is interleaved into high-speed frames on a single link, you can use this interactive TDM simulator.


```json
{"component":"LlmGeneratedComponent","props":{"height":"650px","prompt":"Objective: Create an interactive Time Division Multiplexing (TDM) visualizer to demonstrate how multiple input streams are combined into frames on a single high-speed link.\nData State: initialValues = { connections: 8, inputRateKbps: 1, unitSizeBits: 1 }\nStrategy: Standard Layout. Use Anime.js for data flow animation.\nInputs:\n1. Slider for 'Number of Connections (N)' (range 2-8, default 8).\n2. Number input for 'Input Rate per Connection (Kbps)' (default 1).\nBehavior:\n- Visual Layout: Draw 'N' separate input lines on the left, feeding into a 'Multiplexer (MUX)' box in the center. Draw a single thick 'High-Speed Link' exiting the MUX to the right.\n- Animation: Generate colored blocks (representing 1 bit) moving along the input lines toward the MUX. Give each input line a distinct color. Inside the MUX, animate these blocks assembling into a 'Frame' (a sequence of the colored blocks). Animate the packed frames moving rapidly down the high-speed link.\n- Real-time Calculations Panel: Below the animation, dynamically display the calculations for the current inputs:\n  1. Input Bit Duration (1 / Input Rate)\n  2. Link Transmission Rate (N * Input Rate)\n  3. Time Slot Duration (1 / Link Rate)\n  4. Frame Duration (N * Time Slot Duration)\n- Format the times neatly using ms and μs as appropriate.","id":"im_90da36a8dcef7393"}}
```